



Neural Shifts in Collaborative Team Recommendation

Fani's Lab! @CIKM2025



Fani's Lab/, School of Computer Science, University of Windsor, Canada



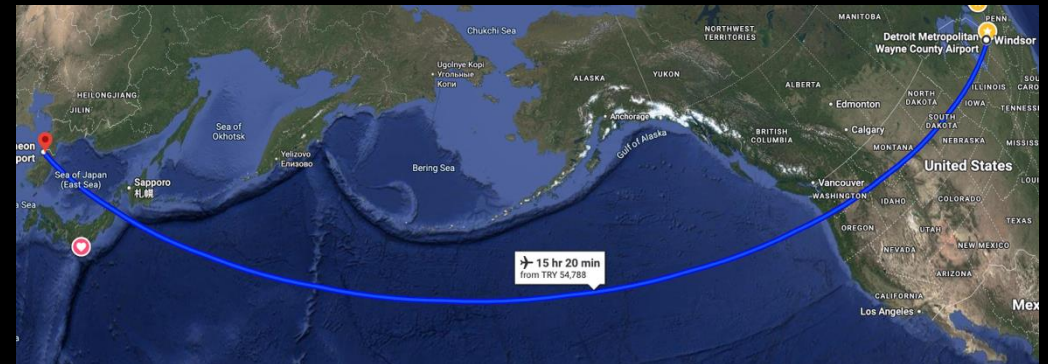
Presenters:

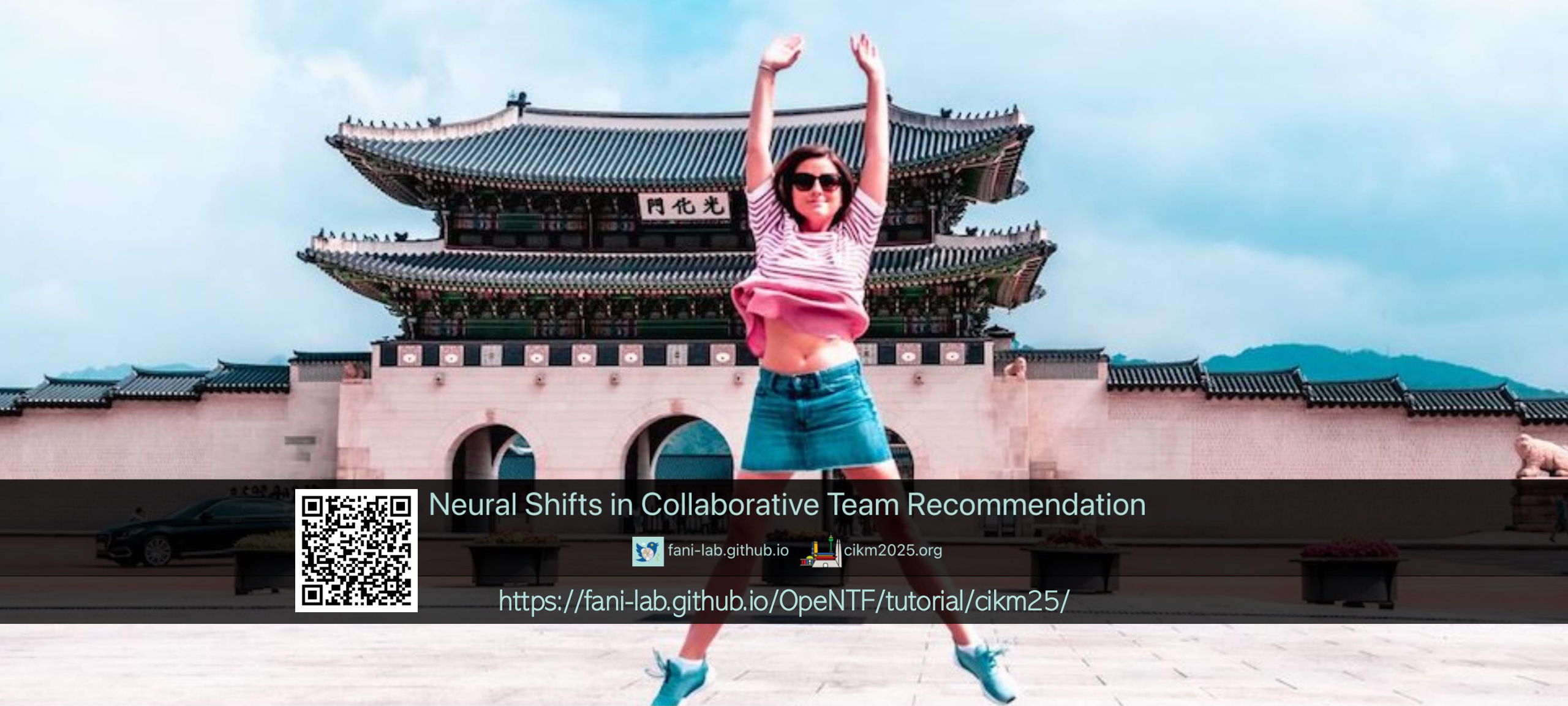


Mahdis Saeedi, PhD., Research Associate



Hossein Fani, PhD., Assistant Professor





Neural Shifts in Collaborative Team Recommendation



fani-lab.github.io



cikm2025.org

<https://fani-lab.github.io/OpeNTF/tutorial/cikm25/>

TIME AND LOCATION

ABSTRACT

TARGET AUDIENCE

OUTLINE

PIONEERING TECHNIQUES

TIME AND LOCATION

Half-day, Morning, 9:00 AM - 12:30 PM (GMT/UTC +9)

Monday, November 10, 2025

COEX, Gangnam, Seoul, South Korea

Prev Tutorials at [UMAP24](#), [SIGIR-AP24](#) and [WSDM25](#) ◀ [\[Full Outline\]](#) [\[Slides\]](#) [\[Recording\]](#)

Outline

- I) Introduction and Background
- II) Pioneering Techniques
- III) Learning-based Heuristics
- IV) Team Refinement
- V) Challenges and New Perspectives
- VI) Applications

Hands-on: OpeNTF

Outline

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II) Pioneering Techniques

III) Learning-based Heuristics

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VI) Applications

Hands-on: OpeNTF



What Is a Team?

A group of users who **collaborate** together with a **common purpose** in order to accomplish the requirements of a task.

[Brannik et al., Psychology Press, 1997]



What Is a Team?

A group of users who **independently** endeavor to accomplish their individual tasks to reach a **shared goal** or value, while **actively interacting and adapting**.

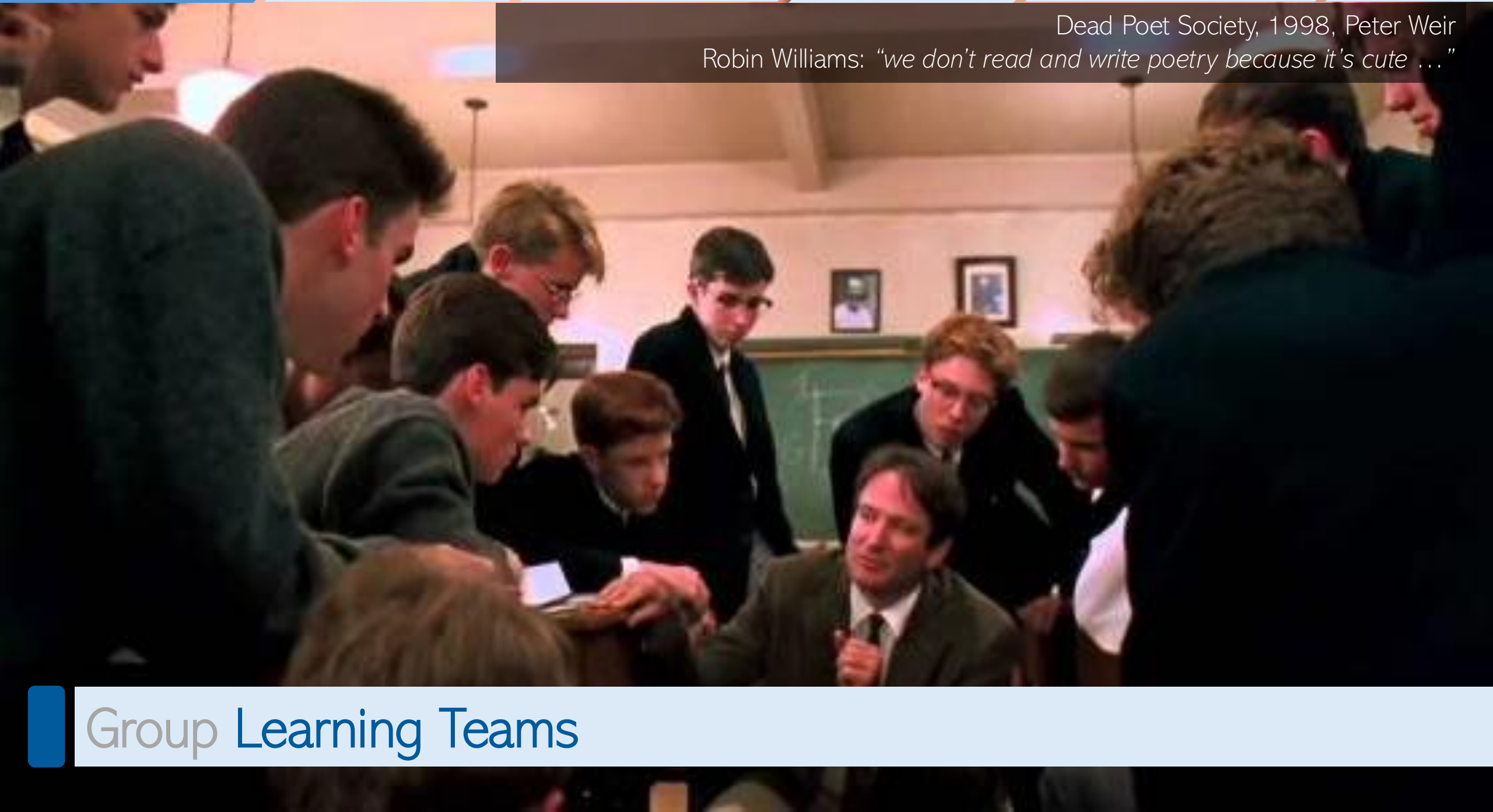
[Zzkarian et. Al., IIE transactions, 1999]



Team vs. Group

Dead Poet Society, 1998, Peter Weir

Robin Williams: *"we don't read and write poetry because it's cute ..."*



Group Learning Teams

Efficient estimation of word representations in vector space

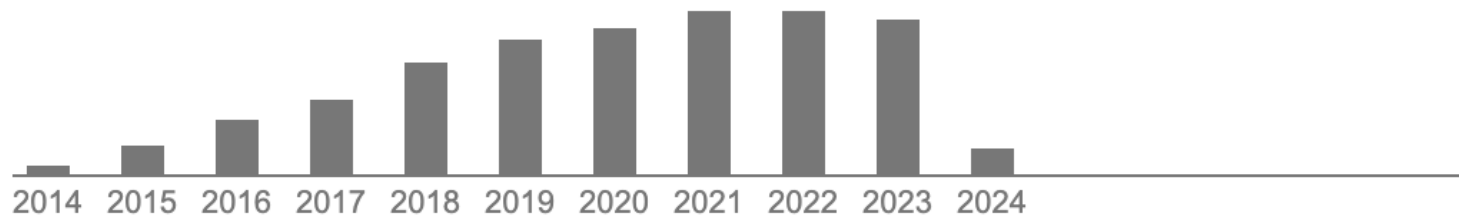
Authors Tomas Mikolov, Kai Chen, Greg Corrado, Jeffrey Dean

Publication date 2013/1/16

Journal arXiv preprint arXiv:1301.3781

Description We propose two novel model architectures for computing continuous vector representations of words from very large data sets. The quality of these representations is measured in a word similarity task, and the results are compared to the previously best performing techniques based on different types of neural networks. We observe large improvements in accuracy at much lower computational cost, i.e. it takes less than a day to learn high quality word vectors from a 1.6 billion words data set. Furthermore, we show that these vectors provide state-of-the-art performance on our test set for measuring syntactic and semantic word similarities.

Total citations Cited by 40332



TITLE	CITED BY	YEAR
Adaptive Cancer Therapy in the Age of Generative Artificial Intelligence Y Derbal Cancer Control 31, 16732746341384704	0	2024
Adaptive control of tumor growth Y Derbal Cancer Control 31, 16732746341320866	1	2024
Cell-Adaptive Fitness and Cancer Evolutionary Dynamics Y Derbal Cancer Informatics. https://doi.org/10.1177/1176838023113465	0	2023
Can artificial intelligence improve cancer treatments? Y Derbal Health Informatics Journal 38(2), 14604582231152219	16	2023
On signaling dysregulation in cancer Y Derbal 2021 International Conference on Computational Science and Computational ...	2	2021
Agent-based model of cell signaling in cancer Y Derbal 2021 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE) ...	1	2020
Modeling Cancer Dynamics Y Derbal 2021 42nd Annual International Conference of the IEEE Engineering in ...		2020
Modeling the Recruitment of Tumor-Infiltrating T Lymphocytes Y Derbal		2019
The adaptive complexity of cancer Y Derbal Bull Math Biol 80(12), 1617220	32	2018
Perspective on the dynamics of cancer Y Derbal Theoretical Biology and Medical Modelling 14, 1-18	22	2017

Science Research Teams

The Big Lebowski, 1998, Joel & Ethan Coen



Entertainment Industry (Cast'nCrew)

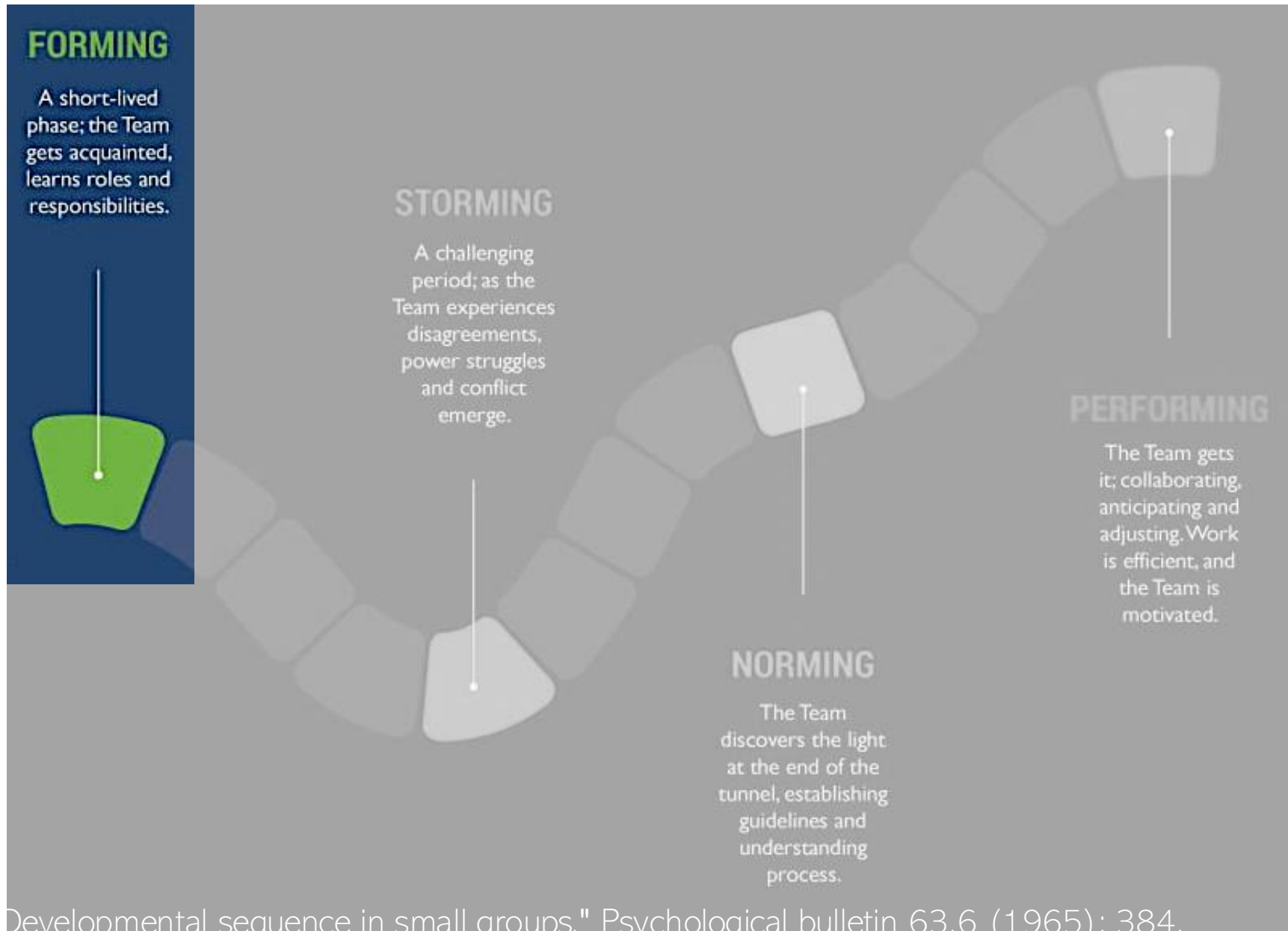


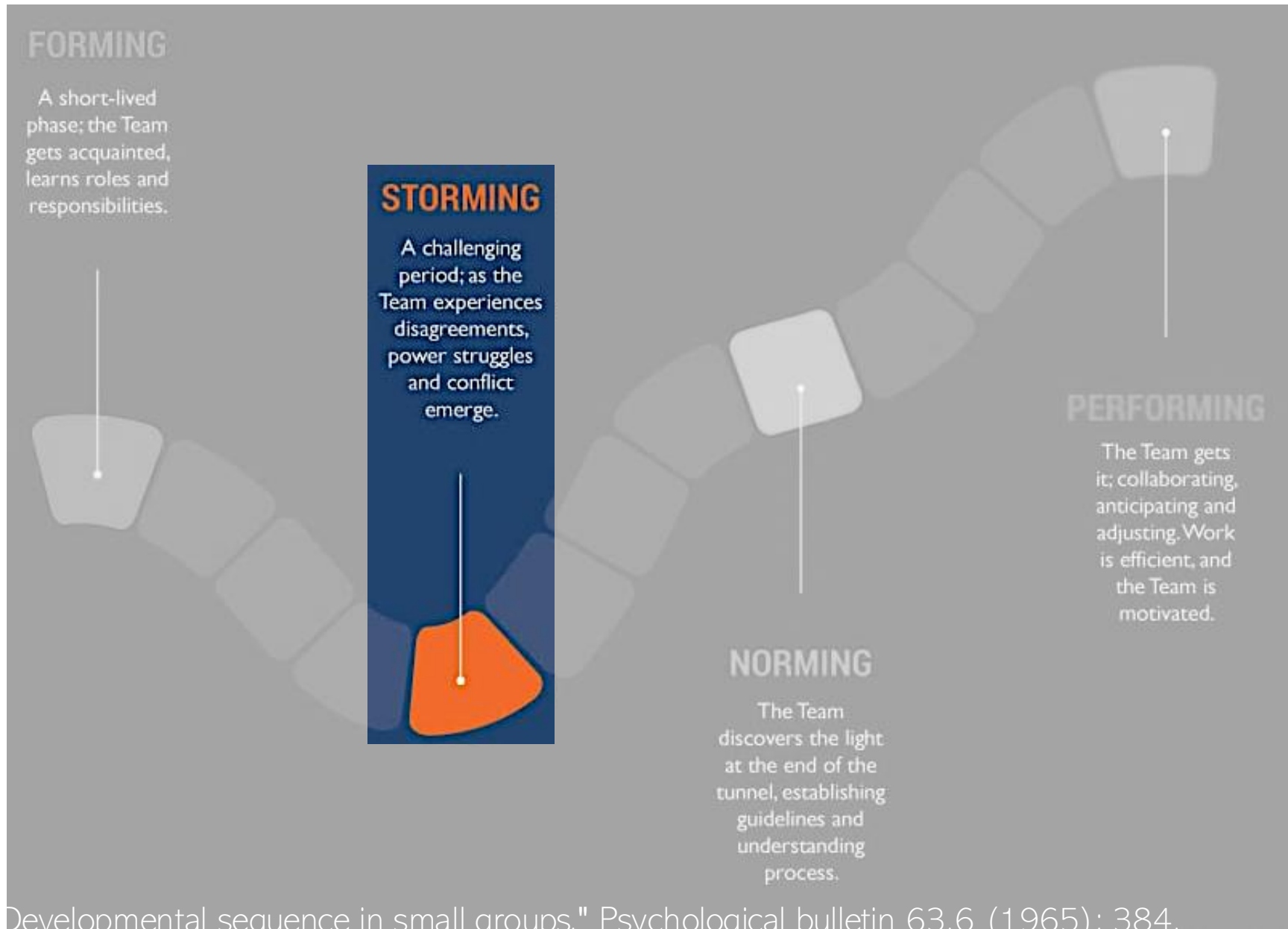
Sport Teams

Development for a Team

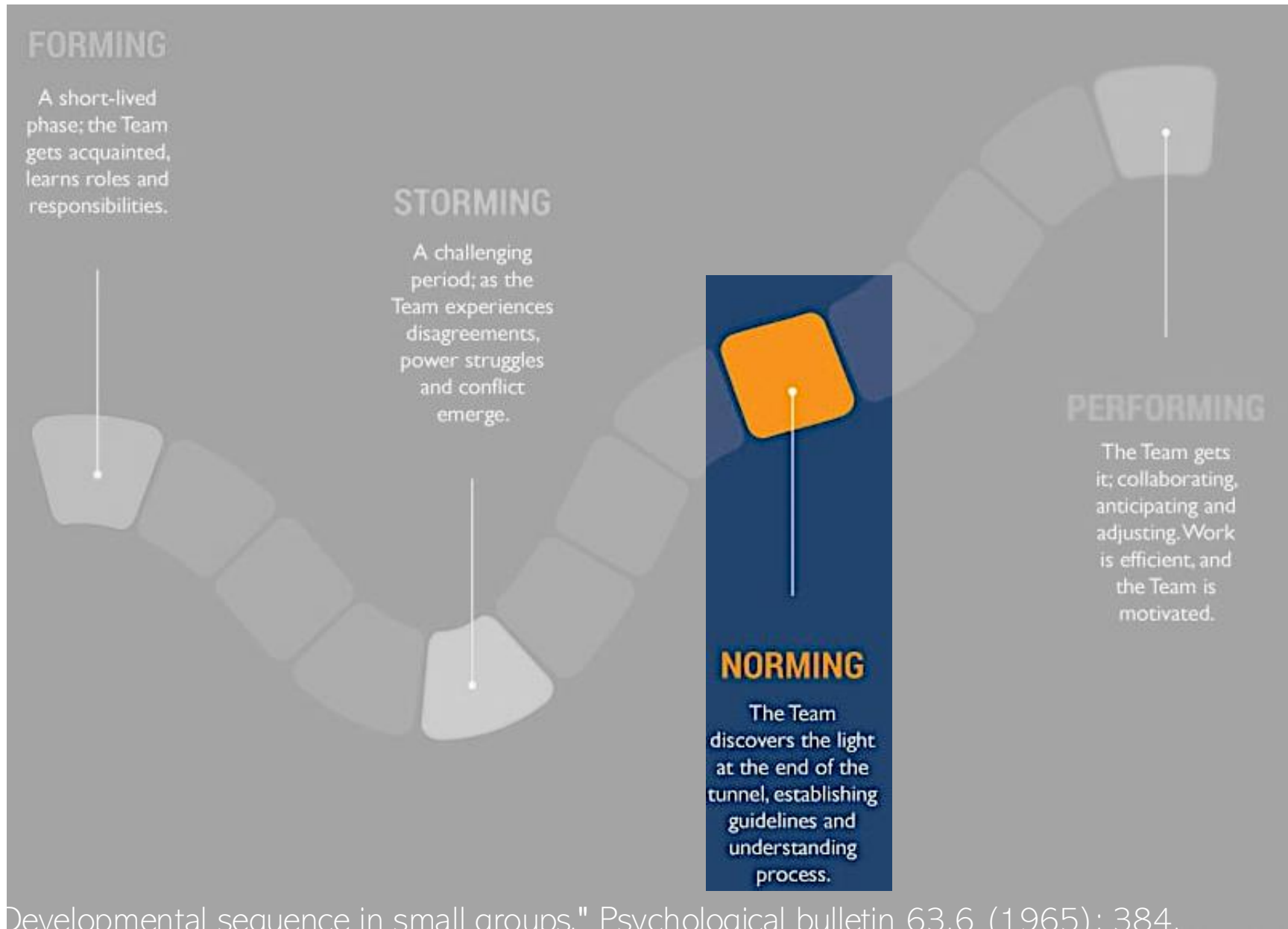
Four distinct phases of development for a team

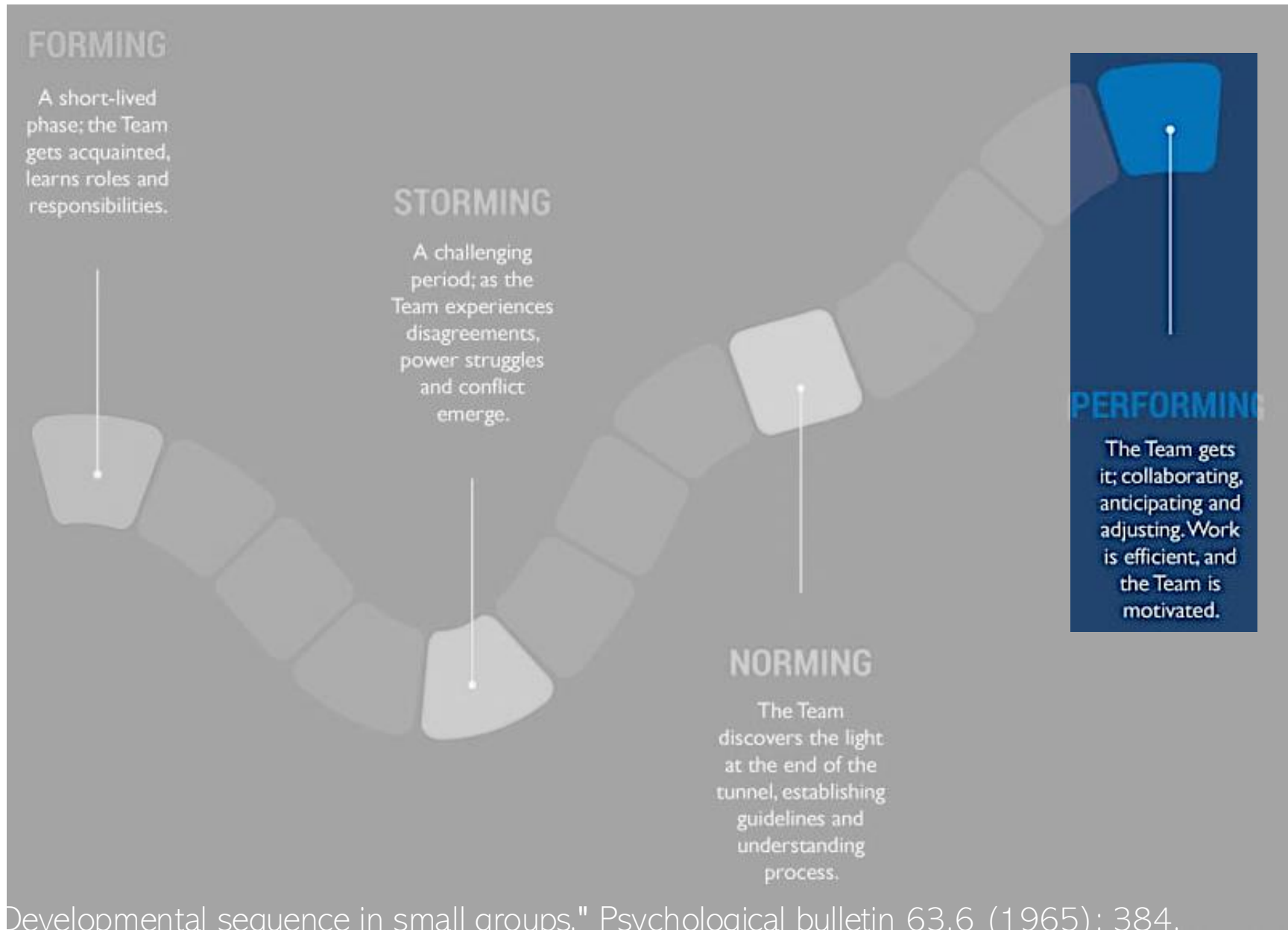
[Tuckman et al., Psychological bulletin ,1965]





Developmental sequence in small groups." Psychological bulletin 63.6 (1965): 384.





Developmental sequence in small groups." Psychological bulletin 63.6 (1965): 384.

Team Recommendation

FORMING

A short-lived phase; the Team gets acquainted, learns roles and responsibilities.





“What does *team recommendation* have to do with *information retrieval*?”



Team Recommendation ↔ Information Retrieval

The anatomy of a large-scale social search engine

Authors:  [Damon Horowitz](#),  [Sepandar D. Kamvar](#) | [Authors Info & Claims](#)

WWW '10: Proceedings of the 19th international conference on World wide web • Pages 431 - 440
<https://doi.org/10.1145/1772690.1772735>

Published: 26 April 2010 [Publication History](#)



👤 297 📈 3,631

Searching the village: models and methods for social search

Authors:  [Damon Horowitz](#),  [Sepandar D. Kamvar](#) | [Authors Info & Claims](#)

Communications of the ACM, Volume 55, Issue 4 • Pages 111 - 118 • <https://doi.org/10.1145/2133806.2133830>

Published: 01 April 2012 [Publication History](#)



👤 13 📈 6,389



Abstract

We present Aardvark, a social search engine that takes web input, text message, or voice. Aardvark uses a social network most likely to be able to answer that question. The challenge lies in finding the right document to satisfy a user's information need, the challenge in a social search engine like Aardvark lies in finding the right person to satisfy a user's information need. Further, while trust in a traditional search engine is based on authority, in a social search engine like Aardvark, trust is based on intimacy. We describe how these considerations inform the architecture, algorithms, and user interface of Aardvark, and how they are reflected in the behavior of Aardvark users.

Abstract

We describe Aardvark, a social search engine. With Aardvark, users ask a question, either by instant message, e-mail, Web input, text message, or voice. Aardvark then routes the question to the person in the user's extended social network most likely to be able to answer that question. As compared to a traditional Web search engine, where the challenge lies in finding the right *document* to satisfy a user's information need, the challenge in a social search engine like Aardvark lies in finding the right *person* to satisfy a user's information need. Further, while trust in a traditional search engine is based on authority, in a social search engine like Aardvark, trust is based on intimacy. We describe how these considerations inform the architecture, algorithms, and user interface of Aardvark, and how they are reflected in the behavior of Aardvark users.



Information Retrieval in a Village: Social Search
Social Information Retrieval (Social IR)



What is Success?

Korea triumphs



Sport Teams

What is success?

US\$1.446 billion vs. no Oscar!





Tomas Mikolov

Efficient estimation of word representations in vector space

Authors Tomas Mikolov, Kai Chen, Greg Corrado, Jeffrey Dean

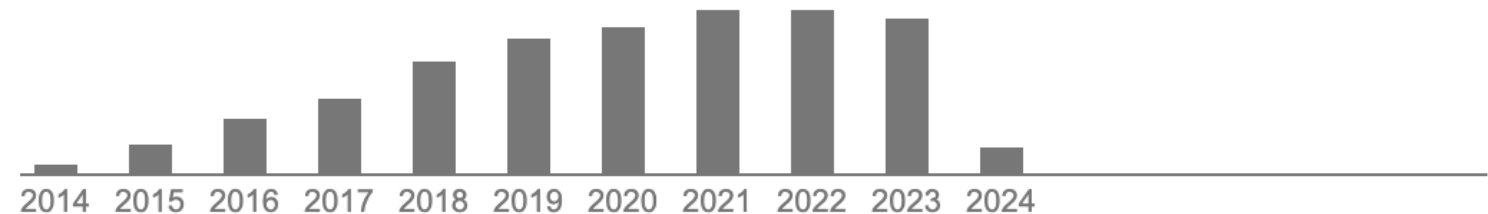
Publication date 2013/1/16

Journal arXiv preprint arXiv:1301.3781

Description We propose two novel model architectures for computing continuous vector representations of words from very large data sets. The quality of these representations is measured in a word similarity task, and the results are compared to the previously best performing techniques based on different types of neural networks. We observe large improvements in accuracy at much lower computational cost, i.e. it takes less than a day to learn high quality word vectors from a 1.6 billion words data set. Furthermore, we show that these vectors provide state-of-the-art performance on our test set for measuring syntactic and semantic word similarities.

Total citations Cited by 40332

What is success?



**Tomas Mikolov**

December 13, 2023 · 🌐

<https://openreview.net/forum?id=idpCdOWtqXd60>

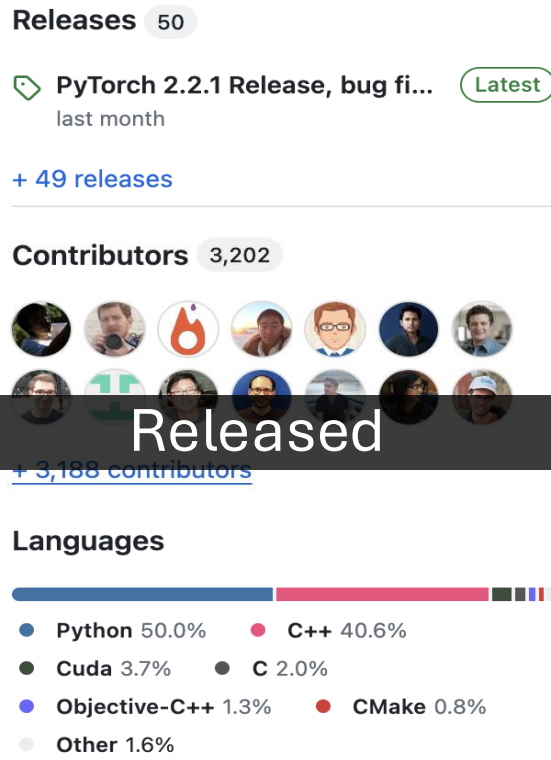
Yesterday we received a **Test of Time Award at NeurIPS** for the word2vec paper from ten years ago. I'm really happy about it! I think it's the first "best paper" type of award I ever received. In fact, the original word2vec paper was **rejected at the first ICLR conference in 2013** (despite the acceptance rate of around 70%), so it made me think how difficult it is for reviewers to predict future impact of research papers.

<https://www.facebook.com/share/p/kXYaYaRvRCr5K2Ze>

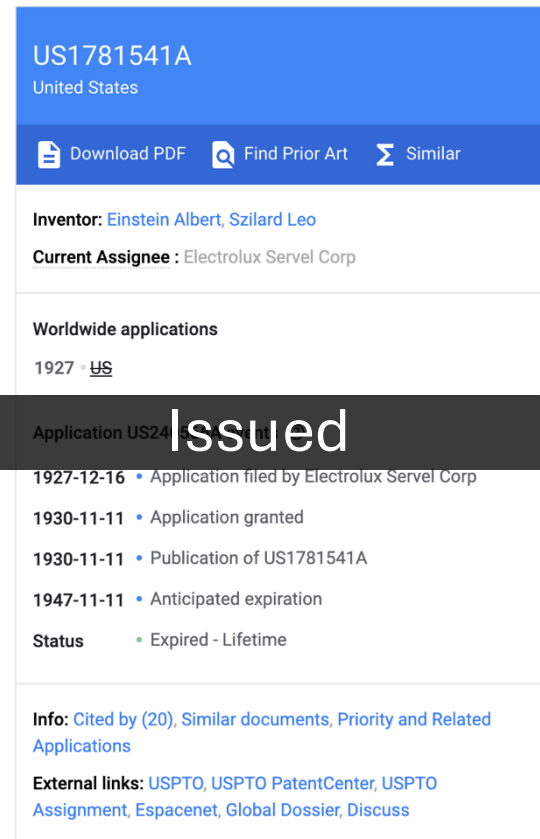
What is success?



Produced



Released



Issued

A Streaming Approach to Neural Team Formation Training

Hossein Fani^[0000–0002–6033–6564], Reza Barzegar^[0009–0002–2831–4143], Arman Dashti^[0000–0001–9022–5403], and Mahdis Saeedi^[0000–0002–6297–3794]

University of Windsor, Windsor, ON., Canada
{hfani, barzegar, vaghehd, msaeedi}@uwindsor.ca

Abstract. Predicting *future* successful teams of experts who can effectively collaborate is challenging due to the experts' temporality of skill sets, levels of expertise, and collaboration ties, which is overlooked by prior work. Specifically, state-of-the-art neural team formation methods learn vector representations for each expert in a static latent space, falling short of incorporating the possible drift and variability of experts' skills and collaboration ties in time. In this paper, we propose (1) a streaming-based training strategy for neural models to capture the evolution of experts' skills and collaboration ties over time and (2) to consume time information as an additional signal to the model for predicting future successful teams. We empirically benchmark our proposed method against state-of-the-art neural team formation methods and a strong temporal recommender system on datasets from varying domains with distinct distributions of skills and experts in teams. The results demonstrate neural models that utilize our proposed training strategy excel at efficacy in terms of classification and information retrieval metrics. The codebase is available at <https://github.com/fani-lab/OpENTF/tree/ecir24>.

Keywords: Neural Team Formation · Training Strategy · OpENTF.

Success

Traditional approach

Teams were formed **manually** by relying on **human experience** and **instinct** in a **tedious**, **error-prone**, and **suboptimal process**.

Difficulties:

- Large number of candidates
 - Different knowledge
 - Different culture
 - Different characteristic
- Hidden personal and societal biases
 - Race
 - Gender
 - Popularity
- Multitude of criteria to optimize
 - Communication cost
 - Budget
 - Time

Traditional approach

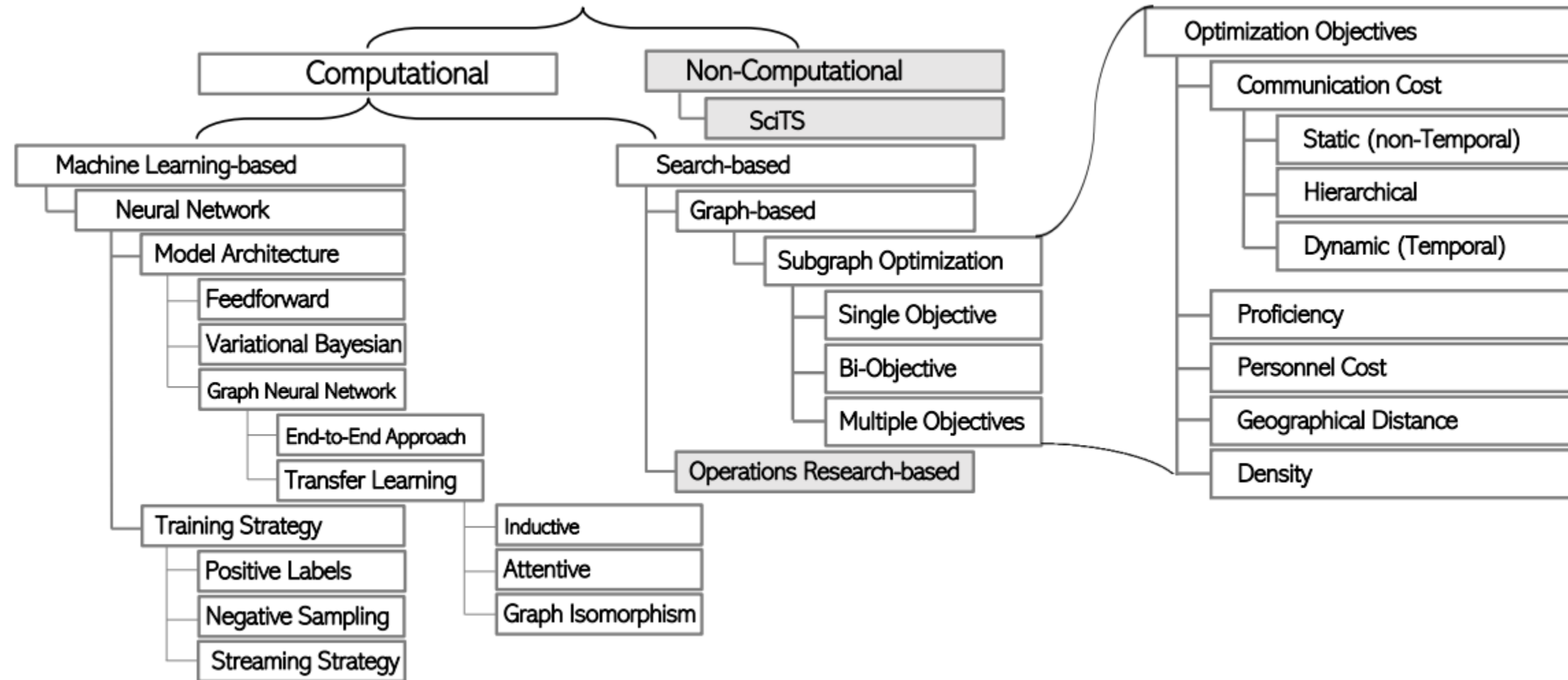
Teams were formed **manually** by relying on **human experience** and **instinct** in a **tedious**, **error-prone**, and **suboptimal process**.

Difficulties:

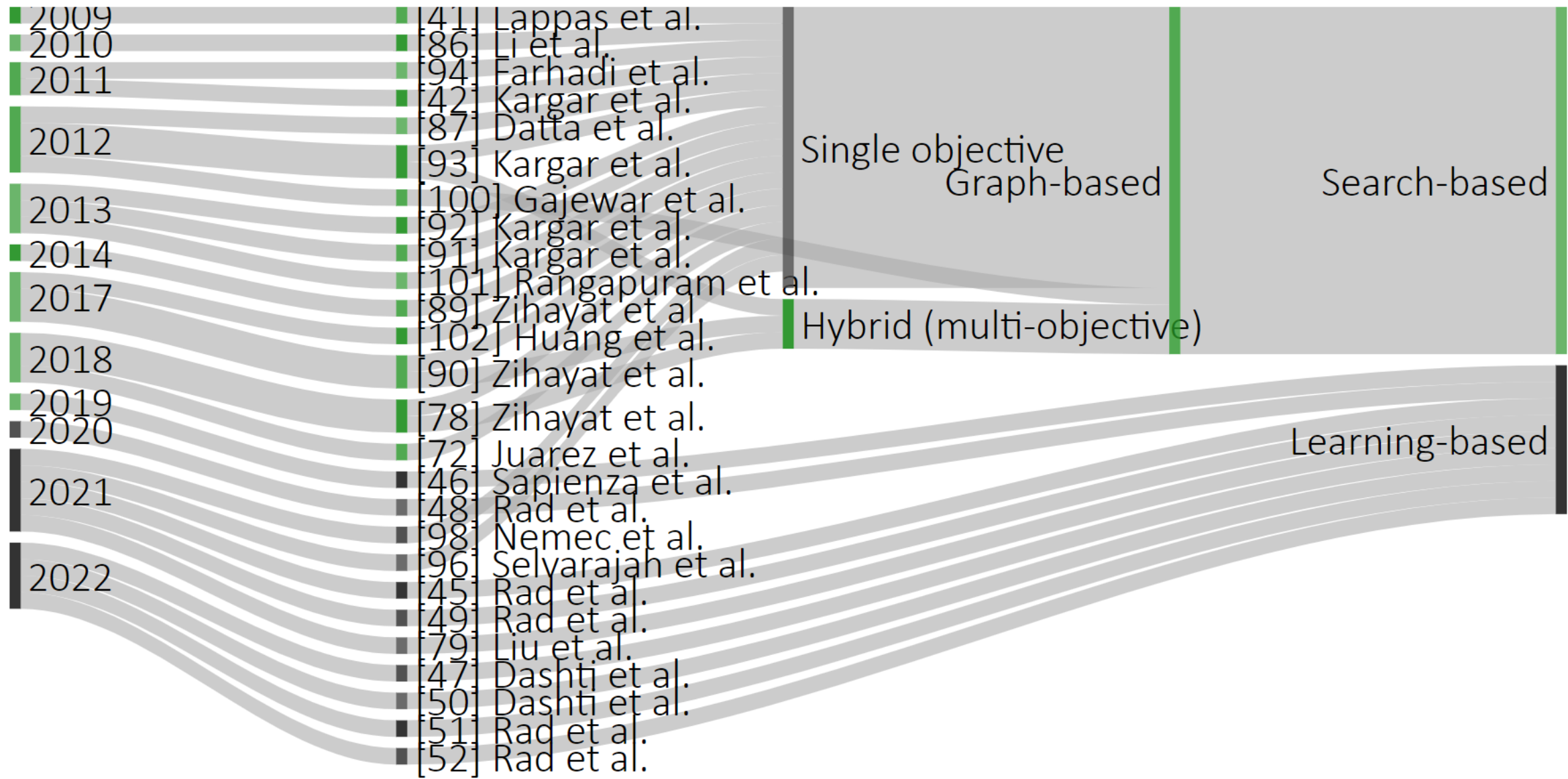
Manual team recommendation on a **large scale** is almost **impossible**

- Race
- Gender
- Popularity
- o Multitude of criteria to optimize
 - Communication cost
 - Budget
 - Time

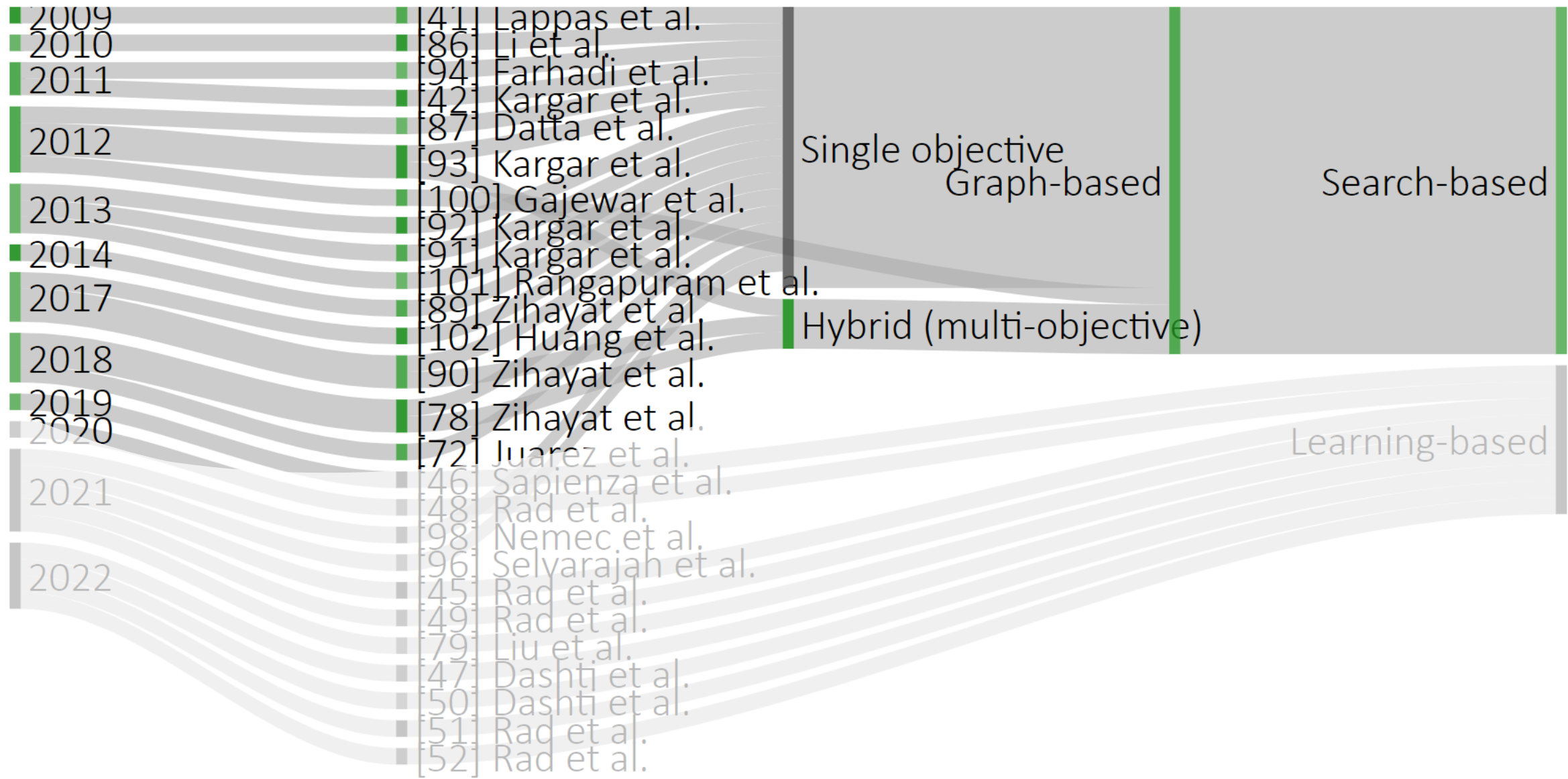
Computational Approach



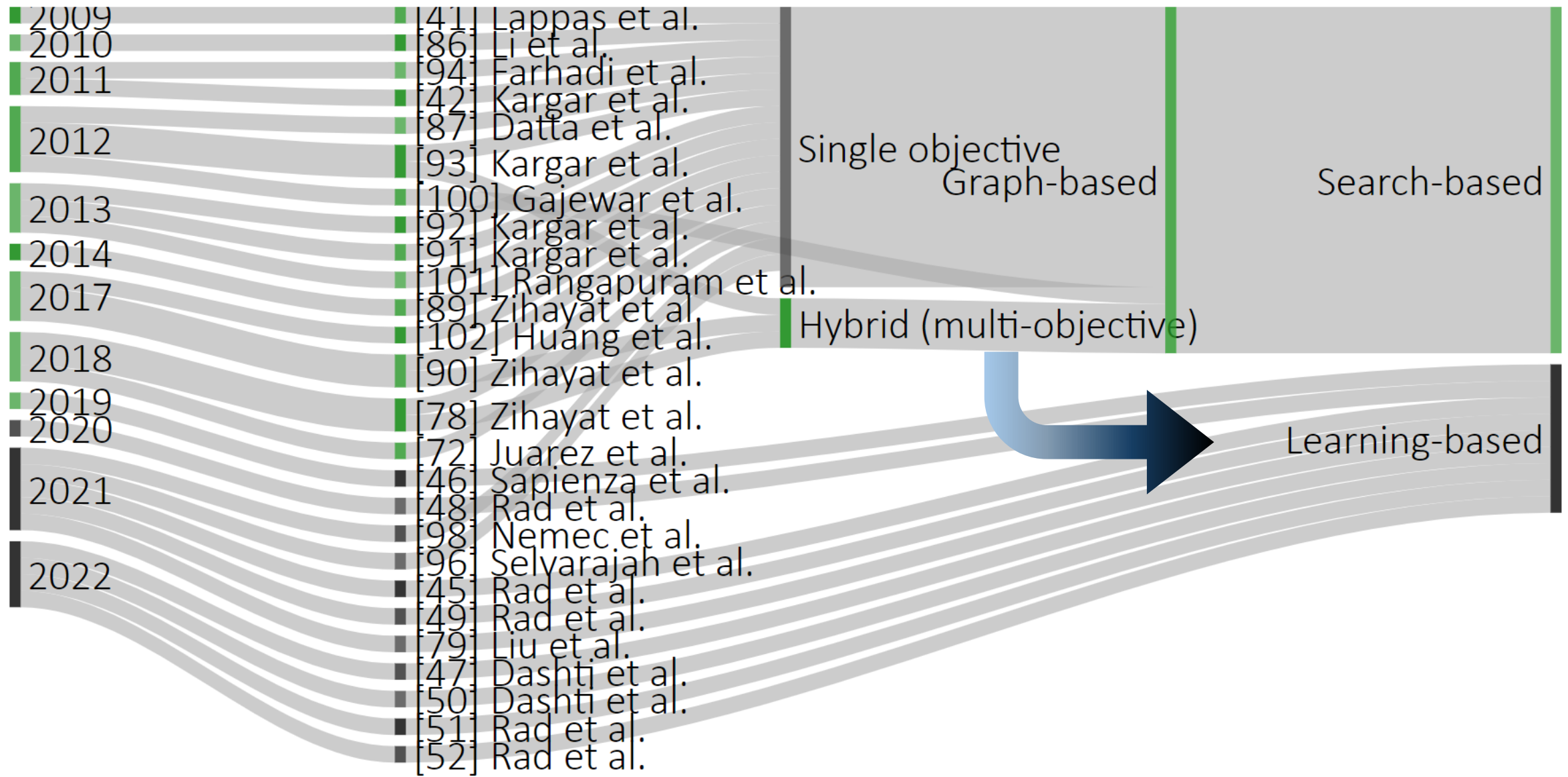
Team Allocation, Team Selection, Team Composition, Team Configuration, Team Recommendation, Team Formation



Team Recommendation Methods in Time



Computationally Prohibitive



Transfer | Distill the Knowledge for Neural Team Recommendation

Outline

I) Introduction and Background

II) Pioneering Techniques

III) Learning-based Heuristics

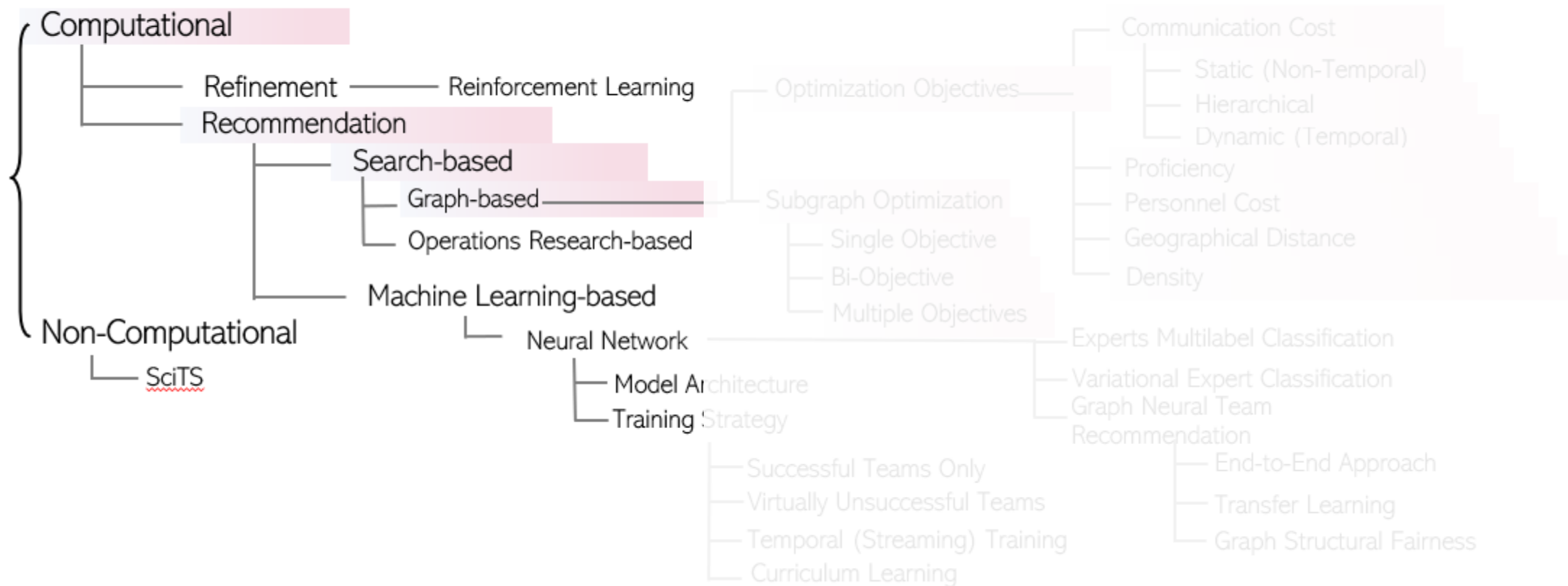
IV) Team Refinement

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Hands-on: OpeNTF

Team Formation Methods



Graph-based Team Recommendation

Underlying **network structure** is a key factor to form an effective team.

[Miller, Computational Modeling and Organization Theories. AAAI Press, 2001]

[Gaston et al., Proceedings of the 1st NAACSOS Conference, 2003]

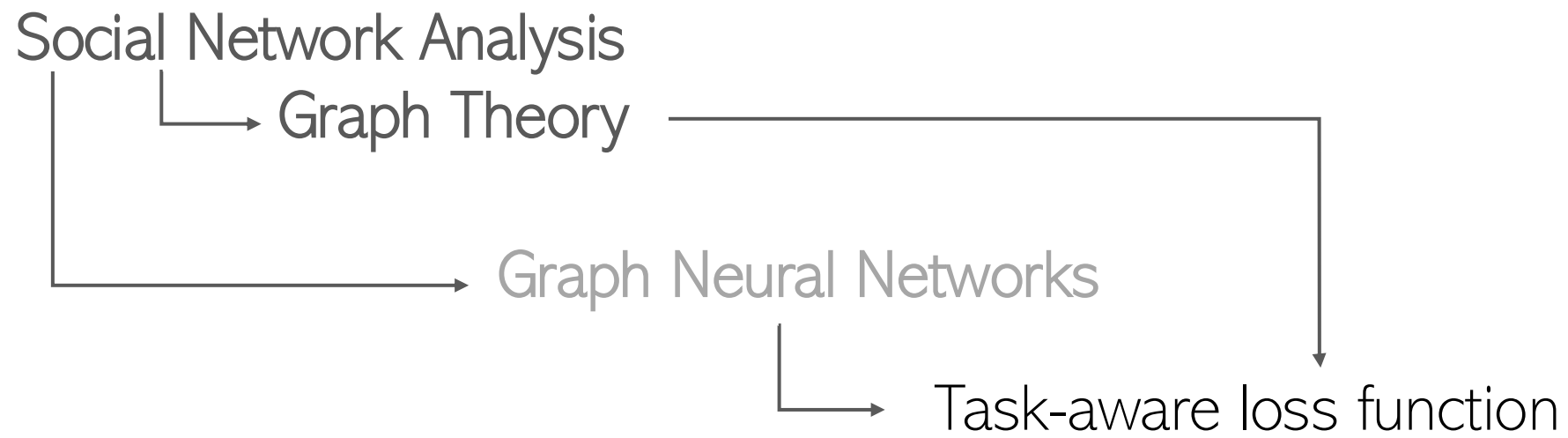
[Gaston et al., AAAI Technical Report, 2004]

[Chen, INCoS, 2010]

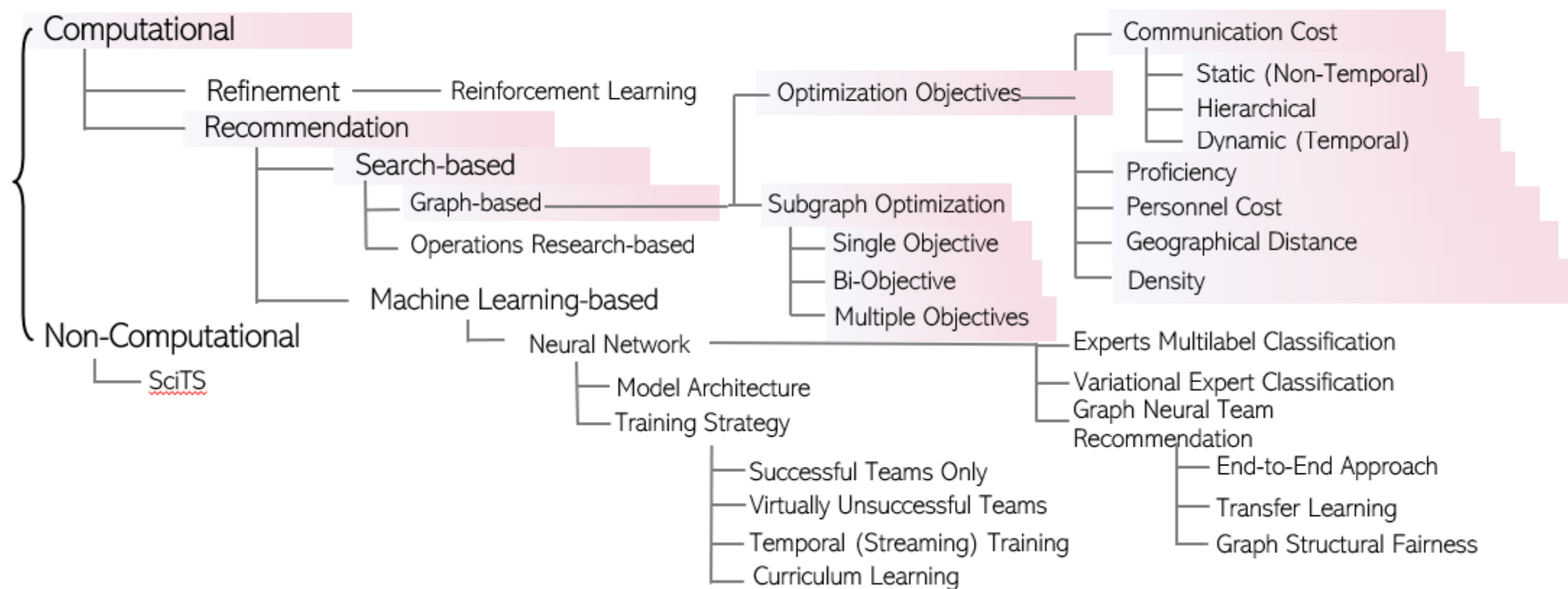
- Organizations' inherent hierarchical structure.
- Experts' social and collaborative ties.

Core Objective

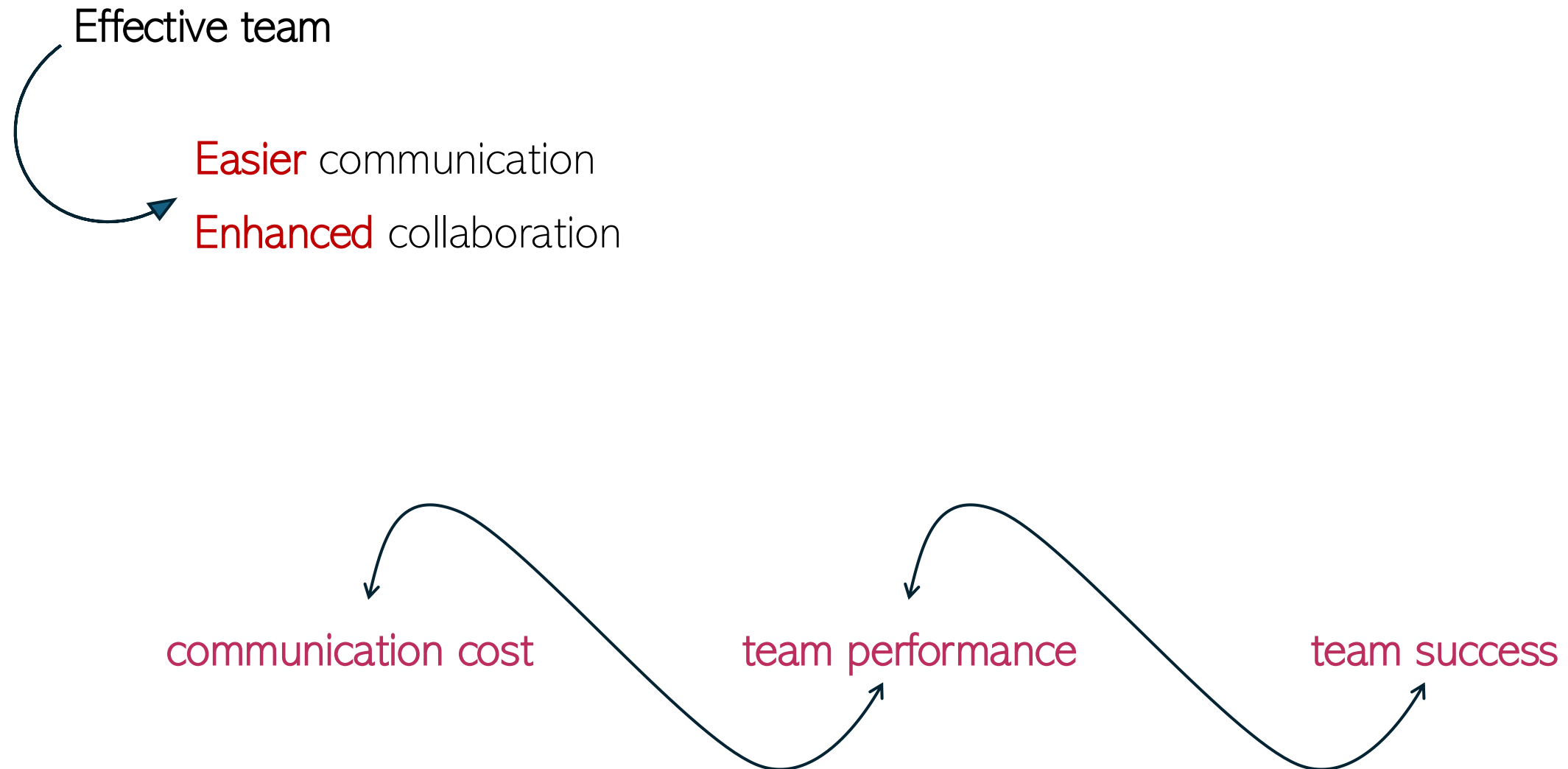
Team Recommendation



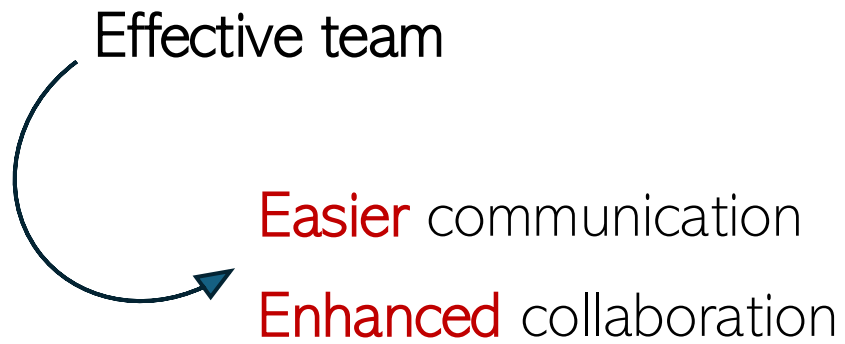
Team Formation Methods



Communication Cost

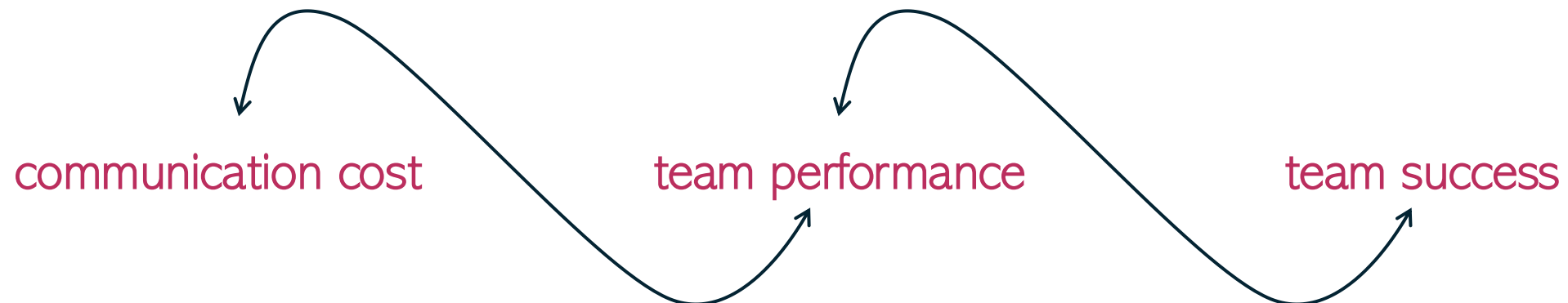


Communication Cost

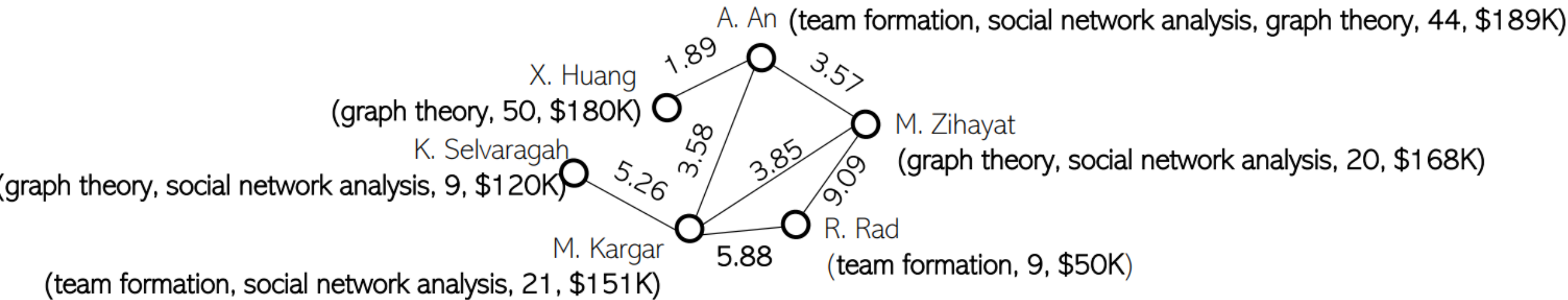


$$W_{\downarrow}(e) = W(u, v) = 1 - \left| \frac{P_u \cap P_v}{P_u \cup P_v} \right| \in R^{[0,1]}$$

$$W_{\downarrow}(e) = \left| \frac{1}{P_u \cap P_v} \right| \in R^{(0,1] \cup \{\infty\}}$$

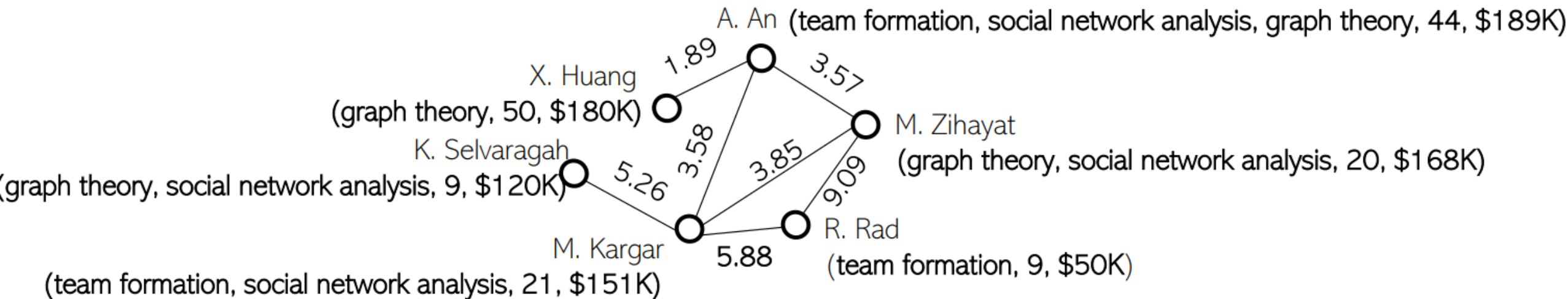


Expert Network $\leftrightarrow$ Attributed Graph



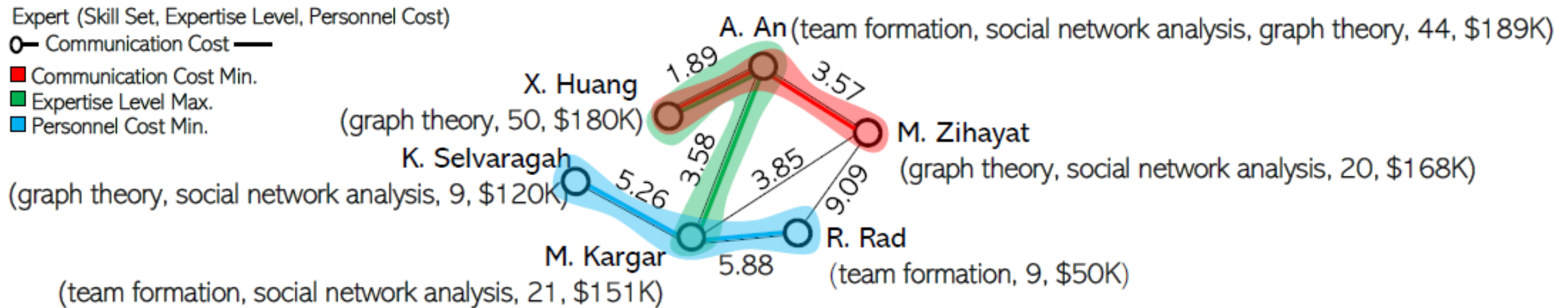
Optimization Objectives

- Communication Cost
- Proficiency
- Personnel Cost
- Geographical Distance
- Density



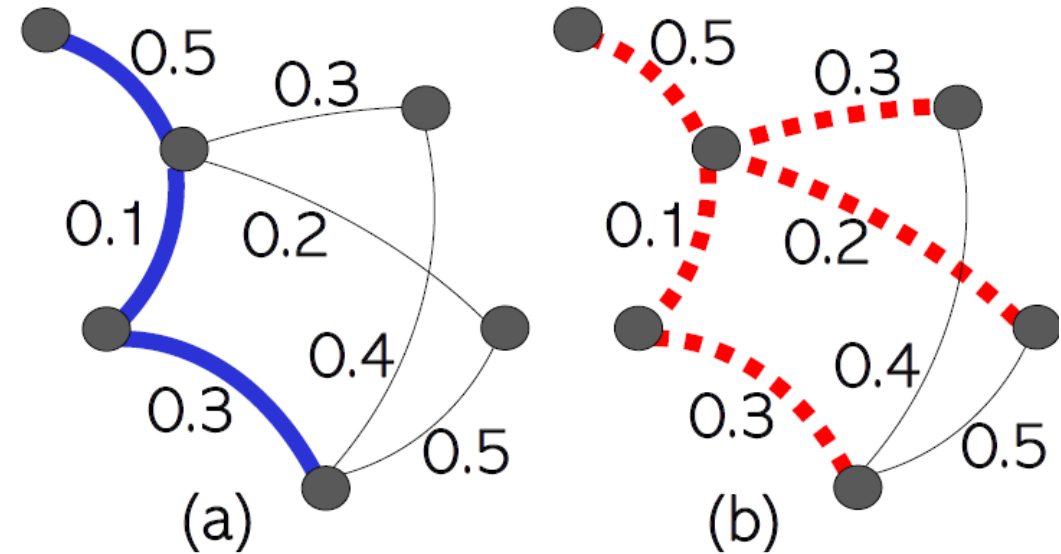
Subgraph Optimization Objectives

- Communication Cost
- Proficiency
- Personnel Cost
- Geographical Distance
- Density



Non-temporal (Static)

- Sum of distances
- Sum of edge weights
- Diameter of the subgraph
- Cost of the spanning tree



Minimum subgraph based on (a) **diameter** (solid blue edges) vs. (b) **spanning tree** (dashed red edges)

	Year	Hybrid	Communication Cost					Proficiency						
			Diameter	MST	Leader Distance	Sum of Edge Weights	Random Walk	Sum of Distance	Trust Score	Expertise Level	Connector Authority	Skill Holder Authority	Personnel Cost	Geographical Distance
Lappas et al. [43]	2009		✓	✓										
Li et al. [92]	2010			✓										
Farhadi et al. [99] (2011)	2011		✓											
Kargar et al. [44]	2011				✓									
Datta et al. [93] (2012)	2012		✓	✓										
Kargar et al. [98]	2012											✓		
Gajewar et al. [105]	2012													✓
Kargar et al. [97]	2013		✓									✓		
Kargar et al. [96]	2013		✓									✓		
Rangapuram et al. [106]	2013												✓	✓
Zihayat et al. [94]	2014								✓			✓		
Huang et al. [107]	2016				✓									
Zihayat et al. [95]	2017					✓					✓	✓		
Zihayat et al. [84]	2018					✓					✓	✓	✓	
Juarez et al. [78]	2018								✓					✓
Nemec et al. [103]	2021						✓							
Selvarajah et al. [101]	2021							✓	✓				✓	

Temporal (Dynamic)

Temporal (dynamic) communication cost is based on the fact that the least communication cost exists between users who could maintain many successful collaborations over time till recently or currently.

$$\min_{V_P \in \mathcal{P}(G)} \varphi_{DT}(G[V_P]) = \sum_{v, v' \in V_P} d(v, v') + \alpha(t - t')$$

Proficiency

Proficiency of users indicates **the level of expertise** in a particular profession or a skill.

- h-index
- number of citations



Jure Leskovec

Professor of Computer Science, [Stanford University](#).
Verified email at cs.stanford.edu - [Homepage](#)

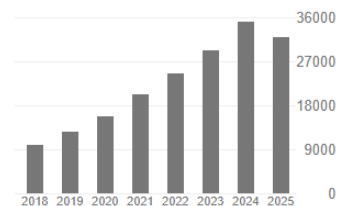
[Machine Learning](#) [AI](#) [Graph Neural Networks](#) [Knowledge Graphs](#) [Complex Networks](#)



TITLE	CITED BY	YEAR
Inductive representation learning on large graphs W Hamilton, Z Ying, J Leskovec Advances in neural information processing systems 30	23032	2017
node2vec: Scalable feature learning for networks A Grover, J Leskovec Proceedings of the 22nd ACM SIGKDD international conference on Knowledge ...	15050	2016
How powerful are graph neural networks? K Xu, W Hu, J Leskovec, S Jegelka arXiv preprint arXiv:1810.00826	12094	2018
On the opportunities and risks of foundation models R Bommasani arXiv preprint arXiv:2108.07258	7405	2021
SNAP Datasets: Stanford large network dataset collection J Leskovec, A Krevl	5237	2014
Graph convolutional neural networks for web-scale recommender systems R Ying, R He, K Chen, P Eksombatchai, WL Hamilton, J Leskovec Proceedings of the 24th ACM SIGKDD international conference on knowledge ...	4917	2018

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not available	available

Based on funding mandates

Proficiency

Proficiency of users indicates **the level of expertise** in a particular profession or a skill.

- h-index
- number of citations

Authority [Zihayat et. Al., EDBT, 2017]

Sum of inverse of expertise level

$$\min_{V_P \in \mathcal{P}(G)} \phi(G[V_P]) = \sum_{v \in A} \frac{1}{a_v}$$



Jure Leskovec

Professor of Computer Science, [Stanford University](#).
Verified email at cs.stanford.edu - [Homepage](#)

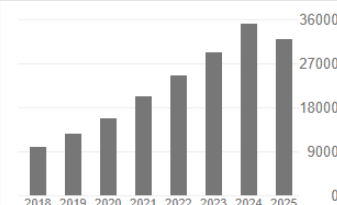
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2 articles	175 articles
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Based on funding mandates

	Year	Hybrid	Communication Cost					Proficiency				Personnel Cost	Geographical Distance	Density	
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Li et al. [92]	2010			✓											
Farhadi et al. [99] (2011)	2011		✓												
Kargar et al. [44]	2011				✓										
Datta et al. [93] (2012)	2012		✓	✓											
Kargar et al. [98]	2012												✓		
Gajewar et al. [105]	2012														✓
Kargar et al. [97]	2013		✓										✓		
Kargar et al. [96]	2013		✓										✓		
Rangapuram et al. [106]	2013													✓	✓
Zihayat et al. [94]	2014								✓				✓		
Huang et al. [107]	2016				✓										
Zihayat et al. [95]	2017					✓					✓	✓			
Zihayat et al. [84]	2018					✓					✓	✓	✓		
Juarez et al. [78]	2018								✓						✓
Nemec et al. [103]	2021						✓								
Selvarajah et al. [101]	2021								✓					✓	

Multi-Objective Optimization

- Communication cost and proficiency

[Zihayat et. al., AMW, 2018]

[Zihayat et. al., EDBT, 2017]

- Communication cost and personnel cost

[Aijun et. al., SDM, 2013]

[Kargar et. al., CSE, 2013]

[Zihayat et. al., AMW, 2018]

- Density and proficiency

[Juarez et. al., GECCO, 2018]

- Communication cost, personnel cost and proficiency

[Zihayat et. al., WI-IAT, 2014]

- Dynamic communication cost, geographical proximity, and proficiency

[Selvarajah et. al., Expert Syst. Appl., 2021]

	Year	Hybrid	Communication Cost					Proficiency				Personnel Cost	Geographical Distance	Density	
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Kargar et al. [44]	2011				✓										
Datta et al. [93] (2012)	2012		✓	✓											
Kargar et al. [98]	2012												✓		
Gajewar et al. [105]	2012														✓
Kargar et al. [97]	2013		✓										✓		
Kargar et al. [96]	2013		✓										✓		
Rangapuram et al. [106]	2013													✓	✓
Zihayat et al. [94]	2014									✓			✓		
Huang et al. [107]	2016				✓										
Zihayat et al. [95]	2017					✓					✓	✓			
Zihayat et al. [84]	2018					✓					✓	✓	✓		
Juarez et al. [78]	2018									✓					✓
Nemec et al. [103]	2021							✓							
Selvarajah et al. [101]	2021								✓	✓				✓	

Subgraph Optimization Algorithms

Subgraph Optimization is **NP-hard**

[Karp, Complexity of computer computations, Springer, 2010]



Heuristics for Polynomial Time

Greedy
Approximation

Communication Cost Minimization Algorithms

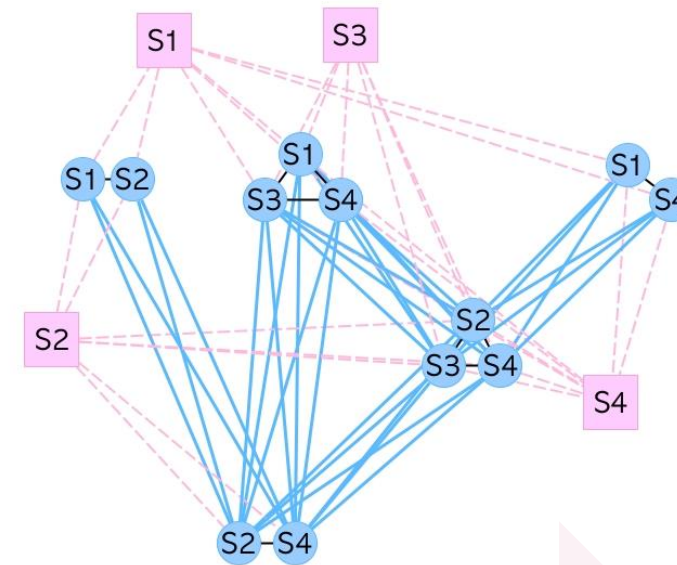
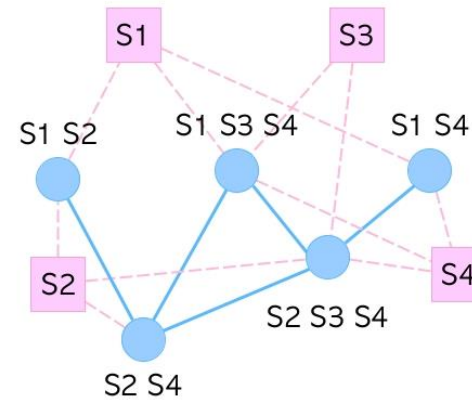
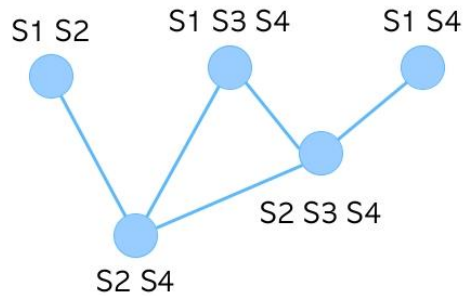
Lappas et al. is the **first attempt** to form a team based on the **subgraph optimization** on the user network. [Lappas et. al., KDD, 2009]

Key Concept: Reducing the cost of communication between team members.

- Diameter-Based Optimization **RarestFirst**
- Spanning Tree-Based Optimization **CoverSteiner** and **EnhancedSteiner**

Communication Cost Minimization Algorithms

Graph extension of EnhancedSteiner Algorithm



A Survey of Subgraph Optimization for Expert Team Formation

ACM Computing Surveys, 2025

Outline

I) Introduction and Background

II) Pioneering Techniques

III) Learning-based Heuristics

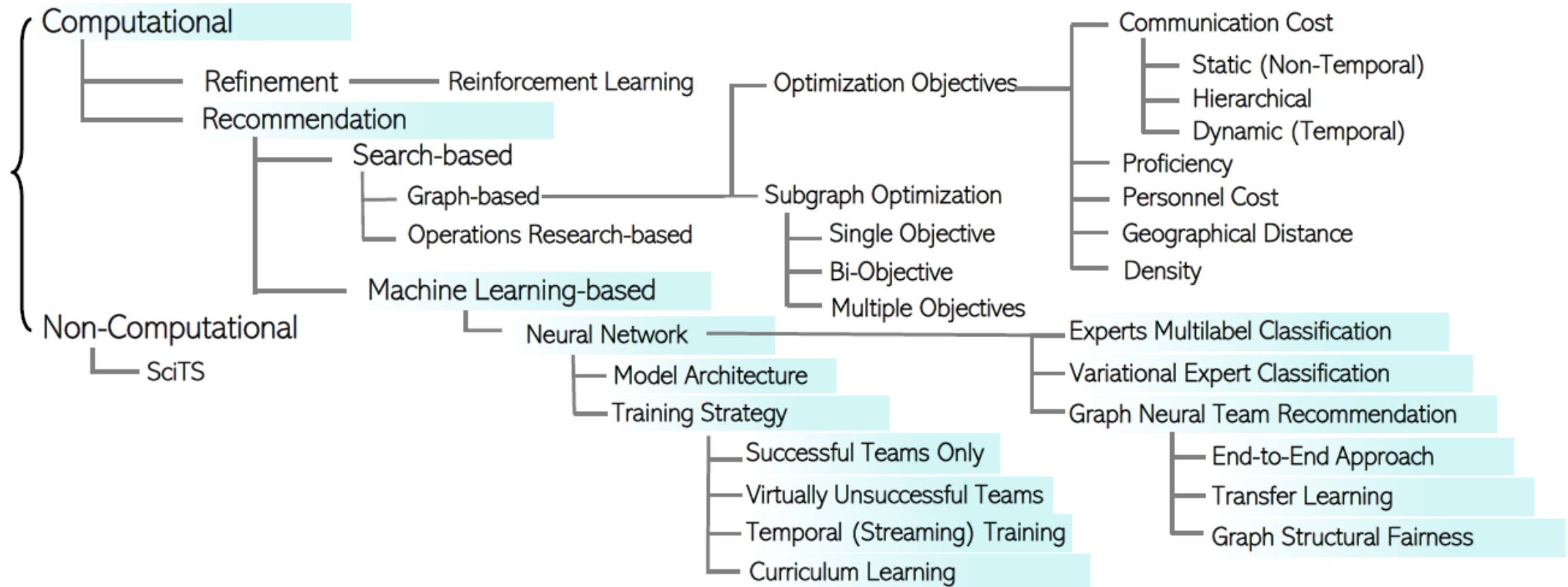
IV) Team Refinement

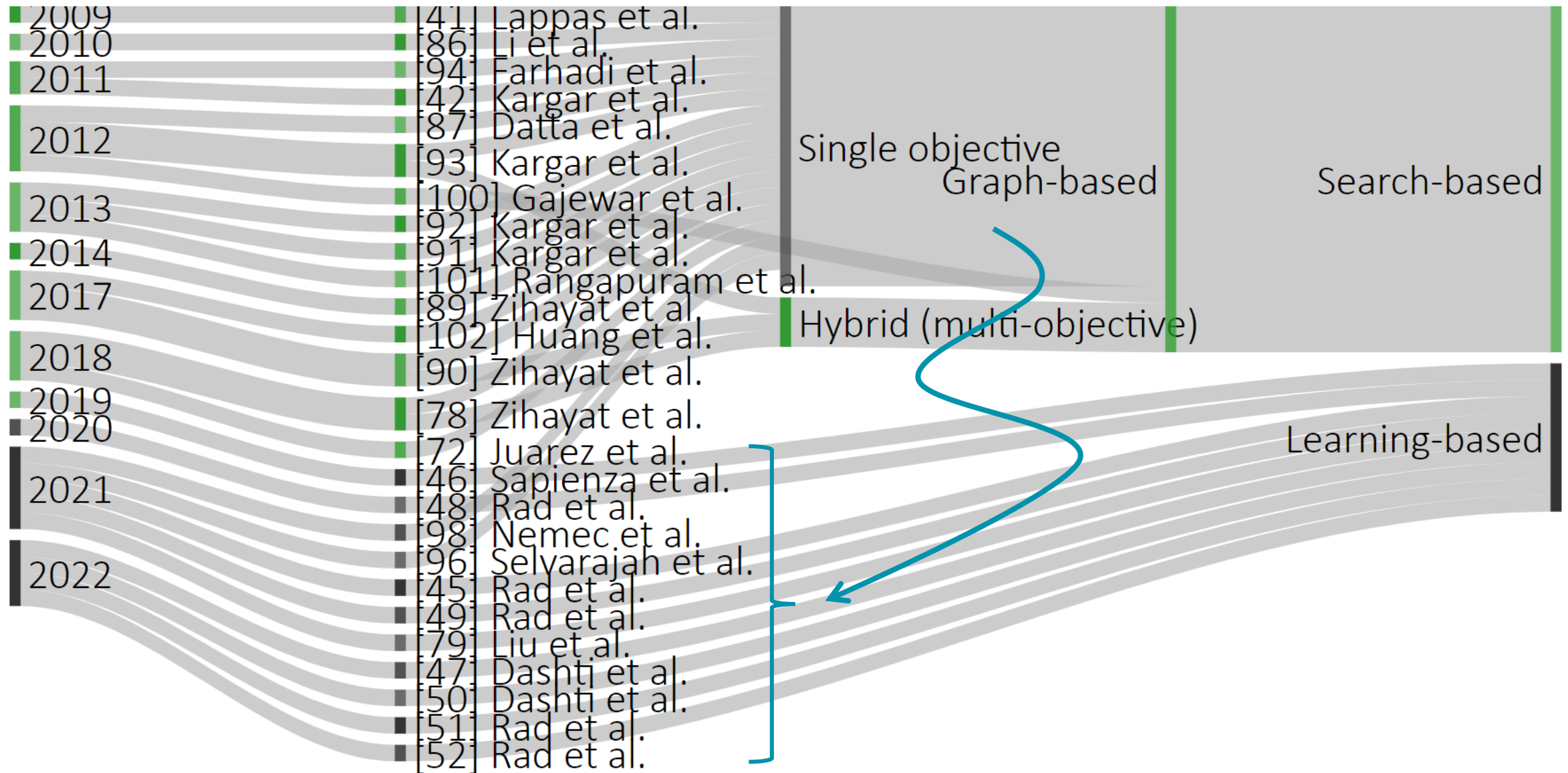
V) Challenges and New Perspectives

VI) Applications

Hands-on: OpeNTF

Team Formation Methods





Paradigm Shift to Learning-based Heuristics



ORIGINAL RESEARCH
published: 13 June 2019
doi: 10.3389/fdata.2019.00014



Deep Neural Networks for Optimal Team Composition

Anna Sapienza[†], Palash Goyal[†] and Emilio Ferrara^{*}

USC Information Sciences Institute, Los Angeles, CA, United States

Cooperation is a fundamental social mechanism, whose effects on human performance have been investigated in several environments. Online games are modern-days natural settings in which cooperation strongly affects human behavior. Every day, millions of players connect and play together in team-based games: the patterns of cooperation can either foster or hinder individual skill learning and performance. This work has three goals: (i) identifying teammates' influence on players' performance in the short and long term, (ii) designing a computational framework to recommend teammates to improve players' performance, and (iii) setting to demonstrate that such improvements can be predicted via deep learning. We leverage a large dataset from Dota 2, a popular Multiplayer Online Battle Arena game. We generate a directed co-play network, whose links' weights depict the effect of teammates on players' performance. Specifically, we propose a measure of network influence that captures skill transfer from player to player over time. We then use such framing to design a recommendation system to suggest new teammates based on a modified deep neural autoencoder and we demonstrate its state-of-the-art recommendation performance. We finally provide insights into skill transfer effects: our experimental results demonstrate that such dynamics can be predicted using deep neural networks.

Keywords: recommendation system, link prediction, deep neural network, graph factorization, multiplayer online games

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Emilio Ferrara

Autoencoder architecture

- **Learn** the inherent structure of ties among users and their skills.
- Utilize all past (un)successful team compositions as training samples.
- **Predict** future teams and their performance.



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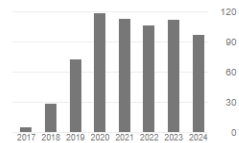
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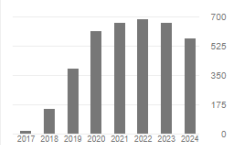
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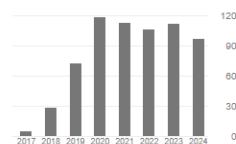
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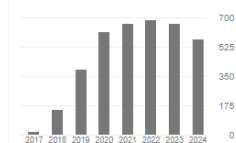
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- Ashok Nish

Neural Team Recommendation

Utilize past (un)successful teams as training samples.

Predict future teams whose success is **almost surely** guaranteed!

Efficiency: iterative and online learning procedures

Efficacy: improved prediction and team performance

Scalability: can handle larger pool of experts

Dynamic Adaptation: better suited for temporal studies

Team

Given a set of **skills** $S = \{i\}$ and a set of **users** $E = \{j\}$, a team of users $e \in E$; $e \neq \emptyset$ that collectively cover the skill set $s \in S$; $s \neq \emptyset$ is shown by (s, e) along with its **success status** $y \in \{0, 1\}$. Further, $T = \{(s, e)_y : y \in \{0, 1\}\}$ indexes all previous teams.

A Streaming Approach to Neural Team Formation Training



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Abstract. Predicting *future* successful teams of experts who can effectively collaborate is challenging due to the experts' temporality of skill sets, levels of expertise, and collaboration ties, which is overlooked by prior work. Specifically, state-of-the-art neural-based methods learn vector representations of experts and skills in a *static* latent space, falling short of incorporating the possible drift and variability of experts' skills and collaboration ties in time. In this paper, we propose (1) a streaming-based training strategy for neural models to capture the evolution of experts' skills and collaboration ties over time and (2) to consume time information as an additional signal to the model for predicting future successful teams. We empirically benchmark our proposed method against state-of-the-art neural team formation methods and a strong temporal recommender system on datasets from varying domains with distinct distributions of skills and experts in teams. The results demonstrate neural models that utilize our proposed training strategy excel at efficacy in terms of classification and information retrieval metrics. The codebase is available at <https://github.com/fani-lab/OpENTF/tree/ecir24>.

Keywords: Neural Team Formation · Training Strategy · OpENTF

Team Recommendation



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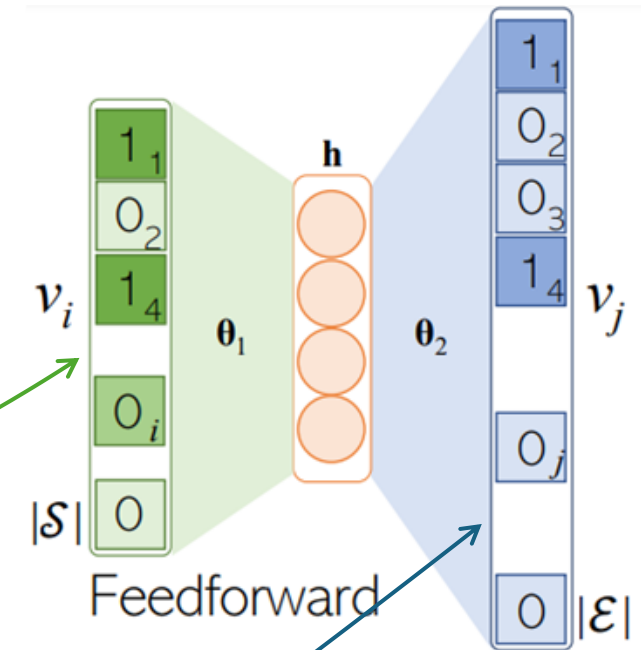
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Given a subset of skills $\mathbf{s}$ and all teams T , the Team Recommendation problem aims at identifying an optimal subset of users $\mathbf{e}^*$ such that their collaboration in the predicted team $(\mathbf{s}, \mathbf{e}^*)$ is successful, that is $(\mathbf{s}, \mathbf{e}^*)_{y=1}$, while avoiding a subset of users $\mathbf{e}'$ resulting in $(\mathbf{s}, \mathbf{e}')_{y=0}$. More concretely, the Team Recommendation problem is to find a mapping function $\mathbf{f}$ of parameters $\boldsymbol{\theta}$ from the powerset of skills to the powerset of $\mathbf{s}$ such that $\mathbf{f}_{\boldsymbol{\theta}} : P(S) \rightarrow P(E)$, $\mathbf{f}_{\boldsymbol{\theta}}(\mathbf{s}) = \mathbf{e}^*$.

Feedforward

Neural Team Recommendation

Given the training set T , Neural Team Recommendation estimates $f_{\theta}(s)$ using a multi-layer neural network that learns, from T , to map a vector representation of subset of skills s , referred to as v_s , to a vector representation of subset of experts e^* , referred to as v_{e^*} , by maximizing a posterior (MAP) probability of θ in f_{θ} over T , that is,

$$\operatorname{argmax}_{\theta} p(\theta | T)$$


$$\begin{aligned}
 h &= \pi(\theta_1 v_s + b_1) \\
 \text{logits} \rightarrow z &= \theta_2 h + b_2 \\
 v_{e^*} &= \sigma(z)
 \end{aligned}$$

Feedforward

Neural Team Recommendation

Biased

$\mathbf{v}_s$, to a vector representation of subset of experts $\mathbf{e}^*$, referred to as $\mathbf{v}_{e^*}$, by maximizing a posterior (MAP) probability of $\boldsymbol{\theta}$ in $f_{\boldsymbol{\theta}}$ over T , that is, $\operatorname{argmax}_{\boldsymbol{\theta}} p(\boldsymbol{\theta} | T)$

$$\begin{aligned} h &= \pi(\boldsymbol{\theta}_1 \mathbf{v}_s + \mathbf{b}_1) \\ \text{logits} \rightarrow \mathbf{z} &= \boldsymbol{\theta}_2 \mathbf{h} + \mathbf{b}_2 \\ \mathbf{v}_{e^*} &= \sigma(\mathbf{z}) \end{aligned}$$

Feedforward → Systematic Discrimination

Traditional FNNs can **overfit** and provide overconfident predictions, especially with **limited** or noisy data because :

- Training sets in team recommendation models suffer from popularity bias;
- The majority of experts (non-popular experts) have scarcely participated in the teams , whereas few experts (popular ones) are in many teams.

Variational Bayesian

Idea:

Using **probabilistic weights** as opposed to single real-valued weights.

- Treating the network's weights as probability distributions instead of fixed values.
- Distribution of predictions rather than a single deterministic output.

Variational Bayesian

Goal:

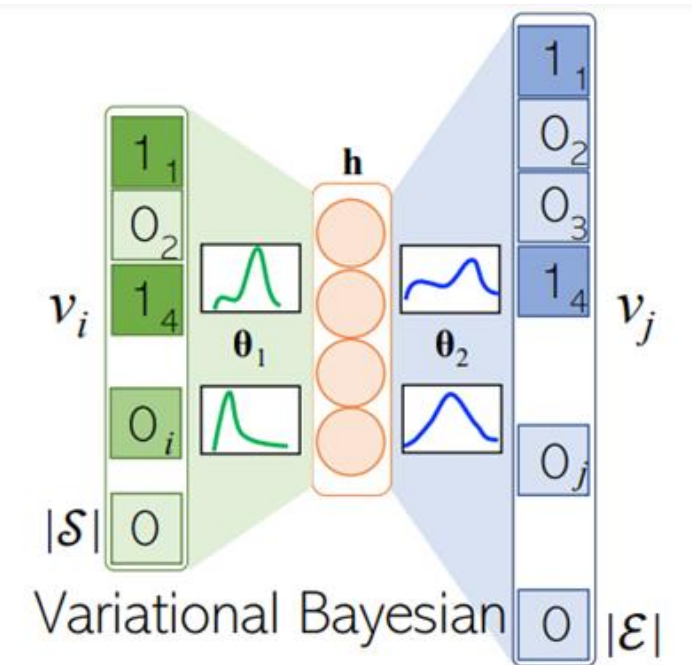
Maximizing the likelihood function f of θ :

- Cross entropy (Classification)
- Sum of squares error (Regression)

Variational Bayesian

By **Bayes theorem**: Multiplying the likelihood with a prior distribution $p(\theta)$.

Hint: The true prior probability of weights $p(\theta)$ **cannot** be calculated efficiently.



$$\theta \sim (\mu, \sigma); q(\theta) = \mathcal{N}(\mu, \sigma) \rightarrow \min \text{KL}(q \parallel p)$$

Minimizing the **Kullback-Leibler Divergence** between $q(\theta)$ and $p(\theta)$

<https://github.com/IntelLabs/bayesian-torch>

Variational Bayesian

By **Bayes theorem**: Multiplying the likelihood with a

Curse of Sparsity at the Input Layer

$$\theta \sim (\mu, \sigma); q(\theta) = \mathcal{N}(\mu, \sigma) \rightarrow \min \text{KL}(q \parallel p)$$

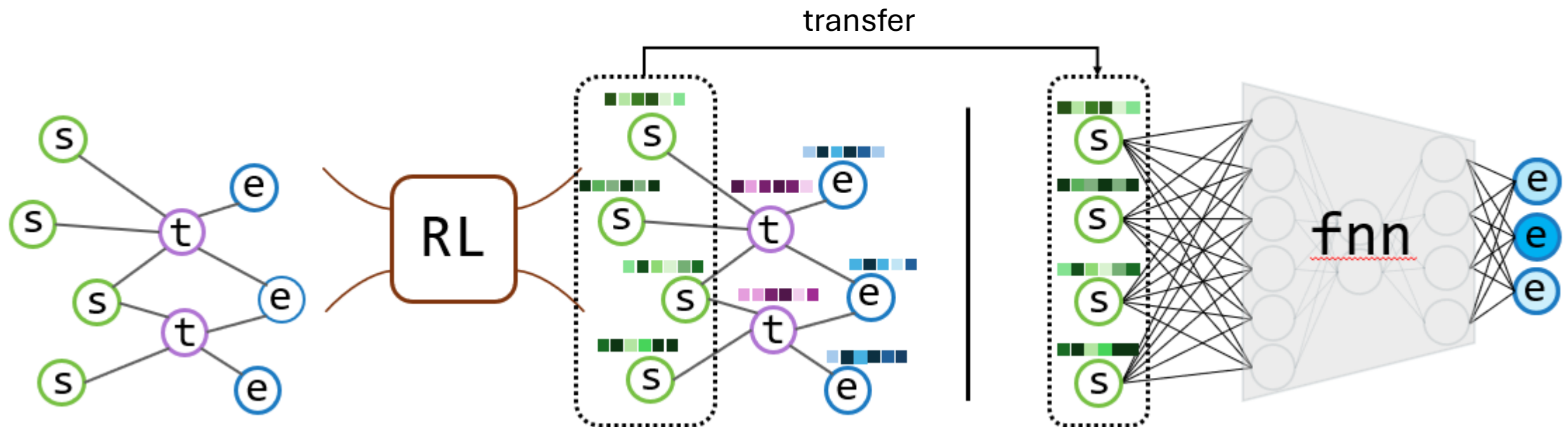
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Variational Bayesian 0 $|\mathcal{E}|$

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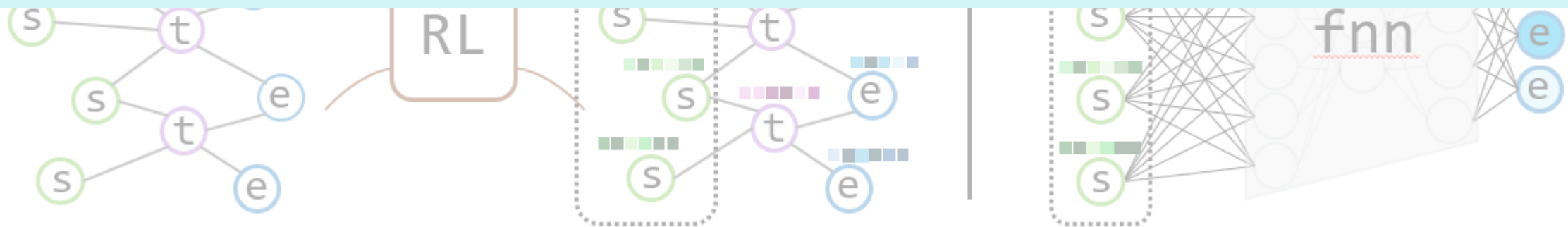
Transfer learning-based



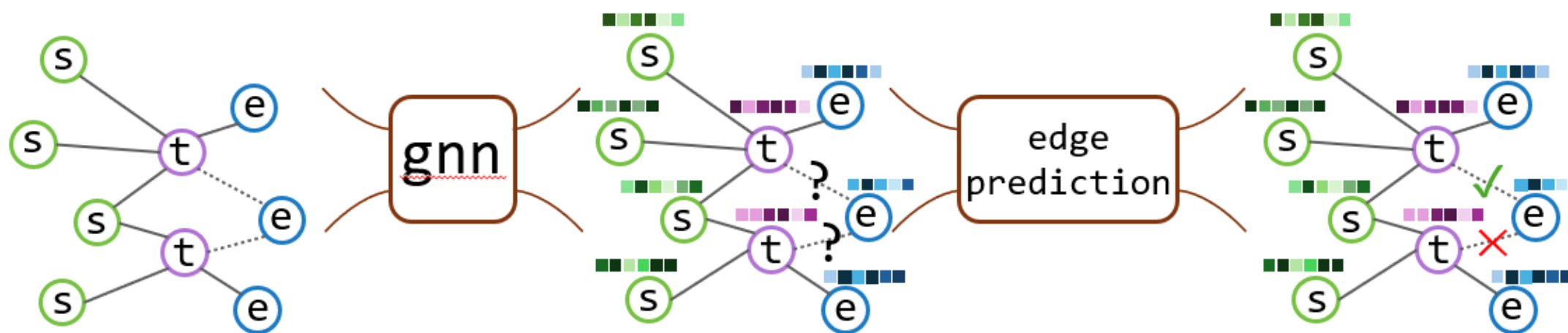
Graph Representation Learning

Transfer learning based

Two step training

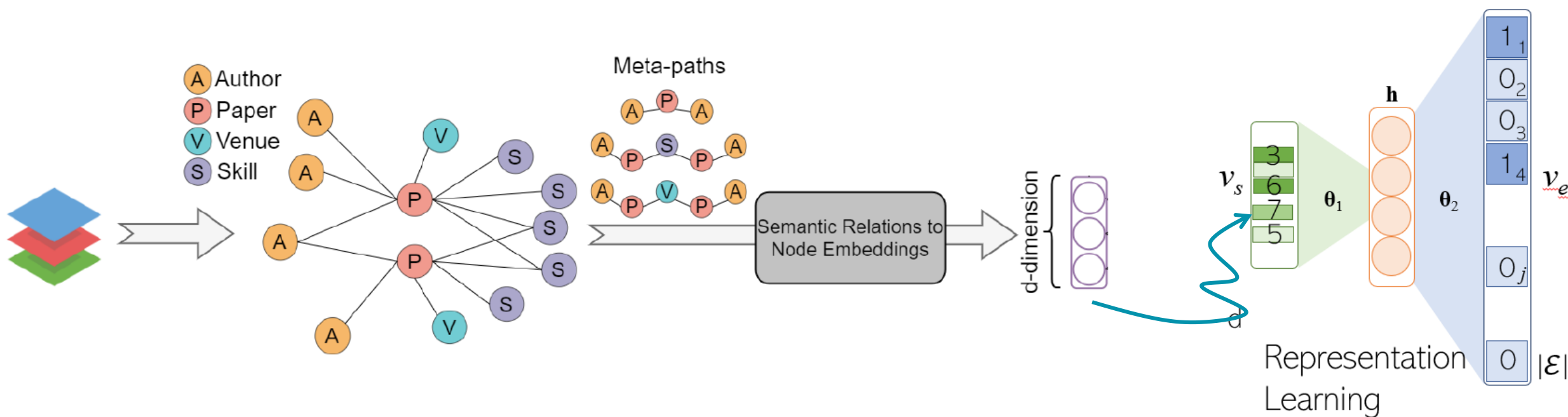


End-to-End Graph Neural Network



Transfer learning-based Methods

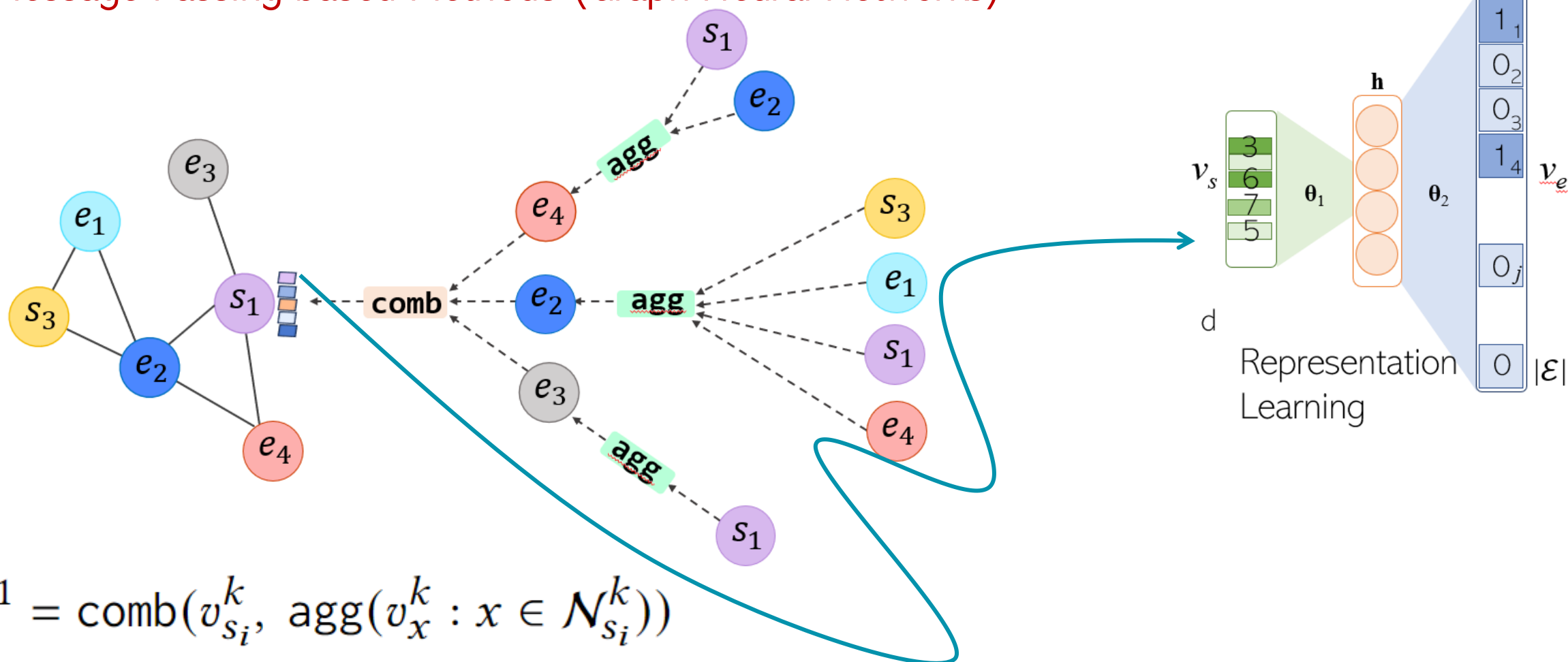
Meta Path-based



[Rad et. Al., Sigir ,2021]

Transfer learning-based Methods

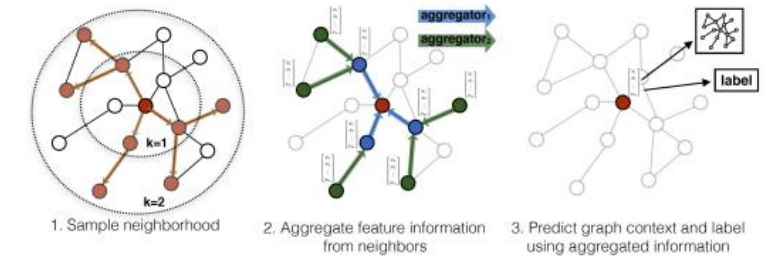
Message Passing-based Methods (Graph Neural Networks)



Graph Representation Learning

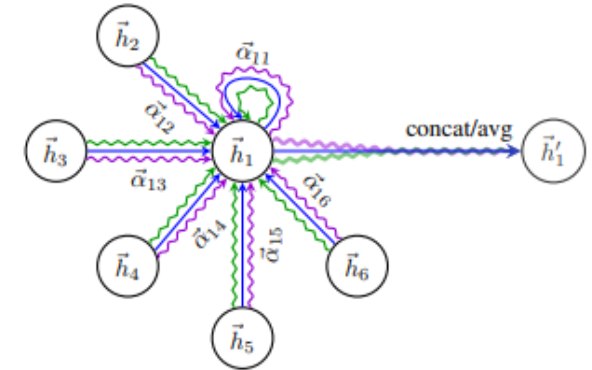
Inductive methods → GraphSAGE

Can infer vector representations for unseen skill nodes based on existing nodes residing in the neighbourhood of the unseen skill node. [Hamilton et. al., NIPS, 2017]



Attentive methods → GAT

Allocate attention coefficients to a skill node's neighbours to signify their importance in updating the skill node's vector. [Velickovic et. al., ICLR, 2018]



Isomorphism methods → GIN

Next to the skill vectors, the agg and comb functions are also parameterized and learnable based on Weisfeiler-Lehman graph isomorphism test. [Xu et al. ICLR 2019]

Graph Neural Network For Team Recommendation

Research Question

Which dense vector representation is of the best quality across neural team recommenders and datasets?

[Ahmed et. Al., WISE ,2024]

[Ahmed et. Al., WISE ,2024]

	%aucroc	%precision			%recall			%ndcg			%map		
		@2	@5	@10	@2	@5	@10	@2	@5	@10	@2	@5	@10
skill-expert													
m-hot [43]	77.54	0.78	0.70	0.65	0.49	1.07	1.96	0.79	0.96	1.37	0.37	0.53	0.66
d2v [16,43,31]	76.36	1.05	0.83	0.72	0.64	1.27	2.21	1.09	1.18	1.62	0.51	0.72	0.88
m2v [38]	76.88	1.21	1.01	0.80	0.73	1.54	2.44	1.25	1.42	1.83	0.58	0.84	0.99
gs [18]	77.81	1.01	0.92	0.83	0.60	1.38	2.51	1.05	1.24	1.76	0.51	0.72	0.90
gat [51]	77.62	1.30	1.10	0.85	0.79	1.66	2.57	1.29	1.49	1.92	0.59	0.84	0.98
gatv2 [5]	77.57	1.28	0.96	0.81	0.78	1.44	2.45	1.33	1.39	1.85	0.62	0.82	0.97
han [53]	77.84	1.39	1.09	0.83	0.84	1.64	2.49	1.46	1.56	1.96	0.67	0.93	1.05
gin [55]	77.25	0.44	0.63	0.69	0.27	0.93	2.07	0.41	0.72	1.25	0.18	0.37	0.53
gine [19]	77.53	1.15	0.92	0.79	0.68	1.37	2.35	1.15	1.26	1.72	0.51	0.71	0.87
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m2v [38]	74.65	0.81	0.74	0.68	0.50	1.14	2.07	0.83	1.01	1.44	0.39	0.59	0.75
gs [18]	77.73	1.12	0.87	0.74	0.67	1.31	2.24	1.19	1.26	1.69	0.55	0.73	0.88
gat [51]	77.87	1.35	1.19	0.92	0.81	1.78	2.76	1.39	1.62	2.08	0.64	0.92	1.09
gatv2 [5]	77.75	1.27	0.92	0.80	0.77	1.39	2.40	1.29	1.32	1.80	0.59	0.77	0.94
han [53]	77.86	1.34	1.07	0.89	0.81	1.61	2.66	1.42	1.53	2.02	0.66	0.90	1.08
gin [55]	77.77	1.31	0.95	0.78	0.78	1.43	2.35	1.35	1.37	1.80	0.61	0.78	0.92
gine [19]	76.70	1.18	1.02	0.83	0.69	1.52	2.49	1.16	1.35	1.80	0.51	0.76	0.92

Average performance of 3-fold fnn on test set in dblp

[Ahmed et. Al., WISE ,2024]

	%aucroc	%precision			%recall			%ndcg			%map		
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gs [18]	75.05	0.70	0.63	0.58	0.42	0.95	1.75	0.71	0.85	1.22	0.32	0.47	0.59
gat [51]	75.41	0.54	0.54	0.53	0.32	0.82	1.60	0.54	0.70	1.06	0.24	0.38	0.49
gatv2 [5]	76.99	1.00	0.87	0.74	0.59	1.32	2.23	1.03	1.20	1.62	0.47	0.68	0.83
han [53]	75.29	0.59	0.52	0.51	0.35	0.79	1.54	0.60	0.71	1.05	0.28	0.40	0.51
gin [55]	72.33	0.80	0.61	0.52	0.48	0.92	1.57	0.83	0.88	1.18	0.40	0.54	0.64
gine [19]	74.10	0.76	0.63	0.56	0.46	0.96	1.71	0.78	0.88	1.23	0.37	0.52	0.63
skill-team-expert													
m-hot [43]	71.42	0.60	0.53	0.49	0.37	0.82	1.51	0.61	0.73	1.05	0.30	0.43	0.54
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gat [51]	76.29	0.72	0.66	0.62	0.44	0.99	1.86	0.72	0.87	1.28	0.33	0.49	0.62
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Average performance of 3-fold bnn on test set in dblp

[Ahmed et. Al., WISE ,2024]

Research Question

Do neural team recommenders of different architectures benefit from dense vectors equally?

[Ahmed et. Al., WISE ,2024]

fnn

	%aucroc	%precision			%recall			%ndcg			%map		
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gat [51]	77.62	<u>1.30</u>	1.10	0.85	<u>0.79</u>	1.66	2.57	1.29	<u>1.49</u>	<u>1.92</u>	0.59	<u>0.84</u>	0.98
gatl2 [5]	77.57	1.28	0.96	0.81	0.78	1.44	2.45	<u>1.33</u>	1.39	1.85	<u>0.62</u>	0.82	0.97
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han [53]	<u>77.86</u>	<u>1.34</u>	<u>1.07</u>	<u>0.89</u>	0.81	<u>1.61</u>	<u>2.66</u>	1.42	<u>1.53</u>	<u>2.02</u>	0.66	<u>0.90</u>	<u>1.08</u>
gin [55]	77.77	1.31	0.95	0.78	0.78	1.43	2.35	1.35	1.37	1.80	0.61	0.78	0.92
gine [19]	76.70	1.18	1.02	0.83	0.69	1.52	2.49	1.16	1.35	1.80	0.51	0.76	0.92

bnn

	%aucroc	%precision			%recall			%ndcg			%map		
		@2	@5	@10	@2	@5	@10	@2	@5	@10	@2	@5	@10
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m2v [38]	70.38	0.46	0.41	0.39	0.29	0.63	1.20	0.47	0.57	0.83	0.22	0.32	0.40
gs [18]	75.05	0.70	0.63	0.58	0.42	0.95	1.75	0.71	0.85	1.22	0.32	0.47	0.59
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han [53]	75.29	0.59	0.52	0.51	0.35	0.79	1.54	0.60	0.71	1.05	0.28	0.40	0.51
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gat [51]	76.29	0.72	0.66	0.62	0.44	0.99	1.86	0.72	0.87	1.28	0.33	0.49	0.62
gatl2 [5]	74.82	0.73	0.64	0.54	0.44	0.96	1.64	0.75	0.87	1.19	0.35	0.49	0.58
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fnn vs bnn performances for dblp

Training Strategies → Negative Sampling

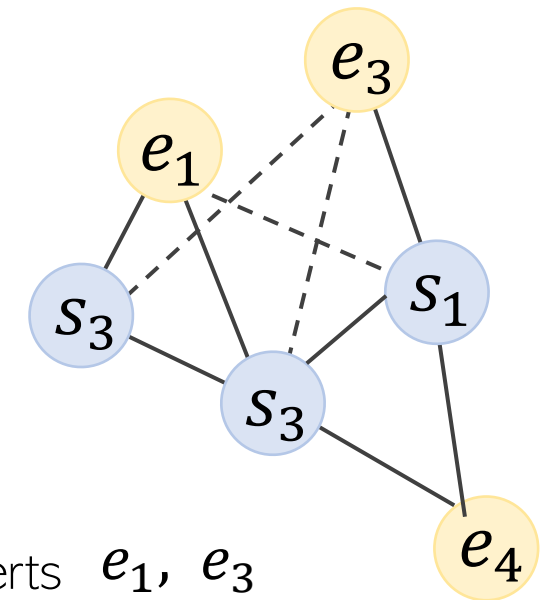
Most available data in team recommendation domain only consists of successful teams.

Closed-World Assumption

No currently known (successful) team



Unsuccessful



Negative team: experts e_1, e_3
skills s_1, s_2, s_3

Training Strategies → Negative Sampling

Optimization function discriminates successful from unsuccessful teams through negative sampling.

Positive Samples

$$\sum_{t_{se} \in \mathcal{T}} [\log \sigma(v_e^\top \cdot v_s) + \sum_{t_{se'} \sim \mathbb{P}: t_{se'} \notin \mathcal{T}}^k \log \sigma(-v_{e'}^\top \cdot v_s)]$$

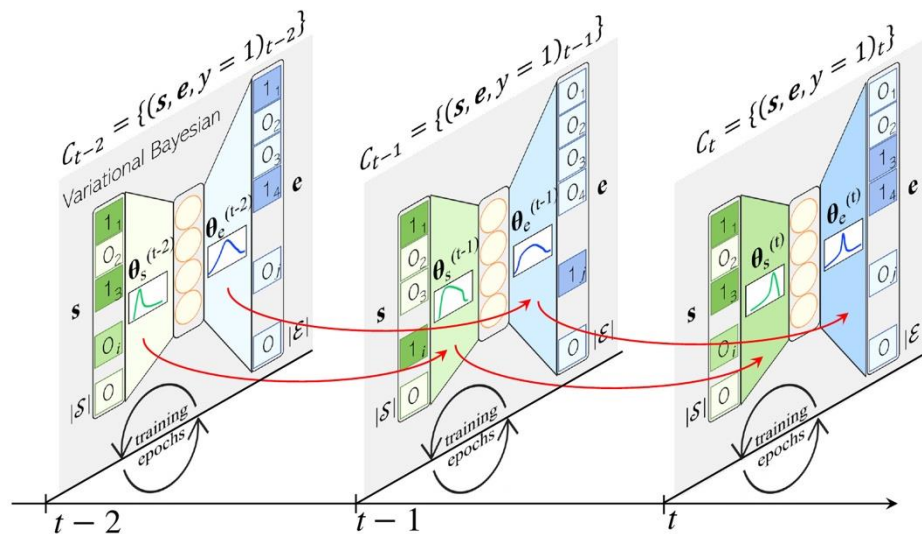
Probability Distribution

Negative Samples

[Dashti et. al., CIKM, 2022]

Training Strategies → Streaming

Predicting future successful teams of users who can effectively collaborate is challenging due to experts' **temporality** of skill sets, levels of expertise and collaboration ties



Neural model learns vector representations for users and skills at time interval t

Initiates learning for the next time interval $t + 1$

Dataset



imdb



Releases 50

 **PyTorch 2.2.1 Release, bug fi...** Latest
last month

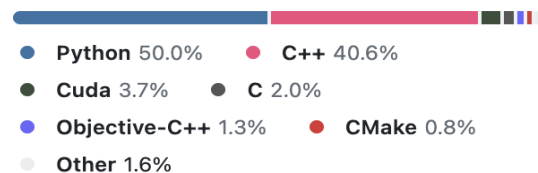
+ 49 releases

Contributors 3,202



+ 3,188 contributors

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Current Assignee: [Electrolux Servel Corp](#)

Worldwide applications
1927-12-16 • Application filed by Electrolux Servel Corp
1930-11-11 • Application granted
1930-11-11 • Publication of US1781541A
1947-11-11 • Anticipated expiration
Status • Expired - Lifetime

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uspt

A Streaming Approach to Neural Team Formation Training

Hossein Fani^[0000-0002-6033-6564], Reza Barzegar^[0009-0002-2831-4143], Arman Dashti^[0000-0001-9022-5403], and Mahdis Saeedi^[0000-0002-6297-3794]

University of Windsor, Windsor, ON., Canada
{hfani, barzegar, vaghehd, msaeedi}@uwindsor.ca

Abstract. Predicting *future* successful teams of experts who can effectively collaborate is challenging due to the experts' temporality of skill

short of incorporating the possible drift and variability of experts' skills and collaboration ties in time. In this paper, we propose (1) a streaming-based training strategy for neural models to capture the evolution of experts' skills and collaboration ties over time and (2) to consume time information as an additional signal to the model for predicting future successful teams. We empirically benchmark our proposed method against state-of-the-art neural team formation methods and a strong temporal recommender system on datasets from varying domains with distinct distributions of skills and experts in teams. The results demonstrate neural models that utilize our proposed training strategy excel at efficacy in terms of classification and information retrieval metrics. The codebase is available at <https://github.com/fani-lab/OpeNTF/tree/ecir24>.

Keywords: Neural Team Formation · Training Strategy · OpeNTF.

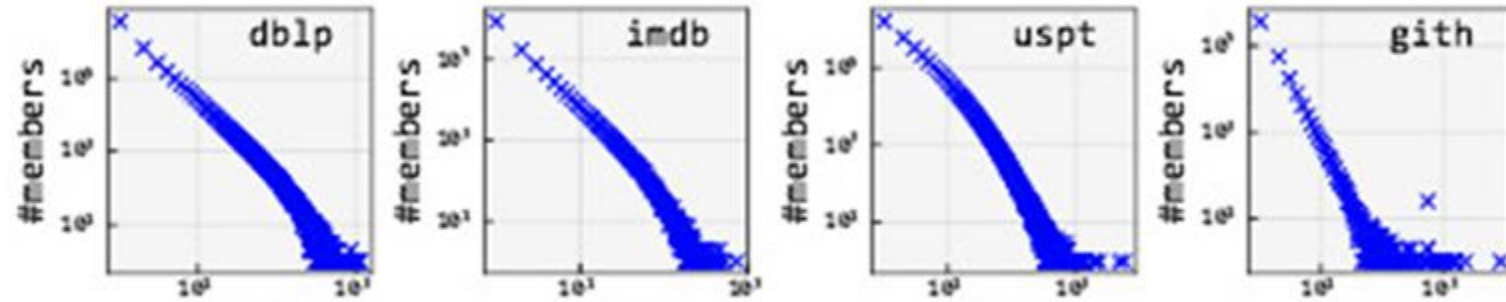
dblp

Dataset

	dblp	uspt	imdb	gith
#teams	4,877,383	7,068,508	507,034	132,851
#unique experts	5,022,955	3,508,807	876,981	452,606
#unique skills	89,504	241,961	28	20
Avg #expert per team	3.06	2.51	1.88	5.52
Avg #skill per team	8.57	6.29	1.54	1.37
Avg #team per expert	2.97	5.05	1.09	1.62
Avg #skill per expert	16.73	19.49	1.59	2.03
#team w/ single expert	768,956	2,578,898	322,918	0
#team w/ single skill	5,569	939,955	315,503	69,131

Dataset Statistics and Distribution

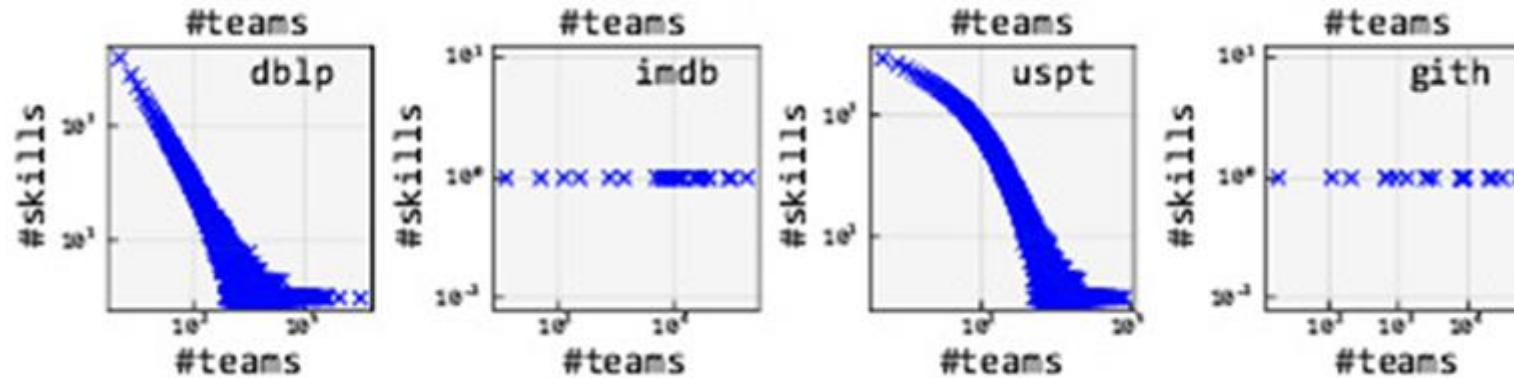
Dataset



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Dataset Statistics and Distribution

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#team w/ single skill	5,569	939,955	315,503	69,131

Dataset Statistics and Distribution

Metrics

Given a test team (s, e)

Precision: How many of the k predicted experts $\hat{e}$ are correctly in e : $P(k) = \frac{|e \cap \hat{e}|}{k}$

Recall: How many of the correct experts e has been predicted in $\hat{e}$: $R(k) = \frac{|e \cap \hat{e}|}{|\hat{e}|}$

Success: Did we at least predict one expert from e : $S(k) = |e \cap \hat{e}| > 0$

Metrics

Reciprocal Rank: the position of first correct expert. The higher rank, the better!

$$RR(k) = \begin{cases} \frac{1}{\text{rank}} & \text{rank} \leq k \\ 0 & \text{otherwise} \end{cases}$$

Average Precision:

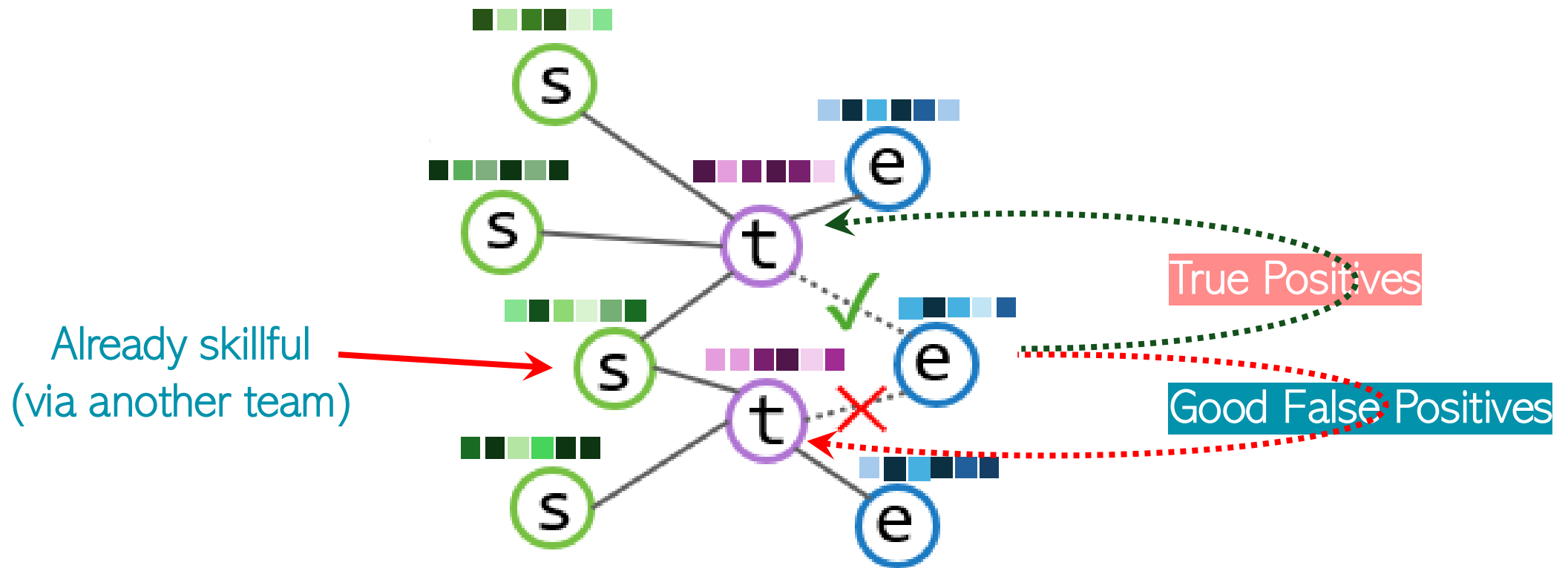
$$AP(k) = \frac{\sum_{i=1}^k P(i) \times \delta_e(i)}{|e|}$$

$\delta_e(i)$ returns 1 if the i -th expert is in e .

Normalized Discounted Cumulative Gain: $nDCG(k) = \frac{\sum_{i=1}^k \frac{\delta_e(i)}{\log(i+1)}}{\sum_{i=1}^{|e|} \frac{1}{\log(i+1)}}$

Metrics

Skill coverage:



Metrics

Metrics		Sapienza <i>et al.</i> [2019]	Rad <i>et al.</i> [2020]	Rad <i>et al.</i> [2021b]	Dashti <i>et al.</i> [2022a]	Rad <i>et al.</i> [2022b]	Rad <i>et al.</i> [2022a]	Dashti <i>et al.</i> [2022b]	Rad <i>et al.</i> [2021a]
+Optimum Teams	Average Precision	✓	✓	✓	✓	✓	✓	✓	✓
	Reciprocal Rank	✓	✓		✓	✓	✓		✓
	Discounted Cumulative Gain	✓	✓	✓	✓	✓	✓	✓	✓
	Recall	✓	✓	✓	✓	✓	✓	✓	✓
	Precision			✓			✓		
	Area Under Curve			✓			✓		
	Squared Error	✓							
	Skill Coverage		✓				✓		
	Communication Cost				✓				

Evaluation Metrics

Outline

I) Introduction and Background

II) Pioneering Techniques

III) Learning-based Heuristics

IV) Team Refinement

V) Challenges and New Perspectives

VI) Applications

Hands-on: OpeNTF

Team Refinement

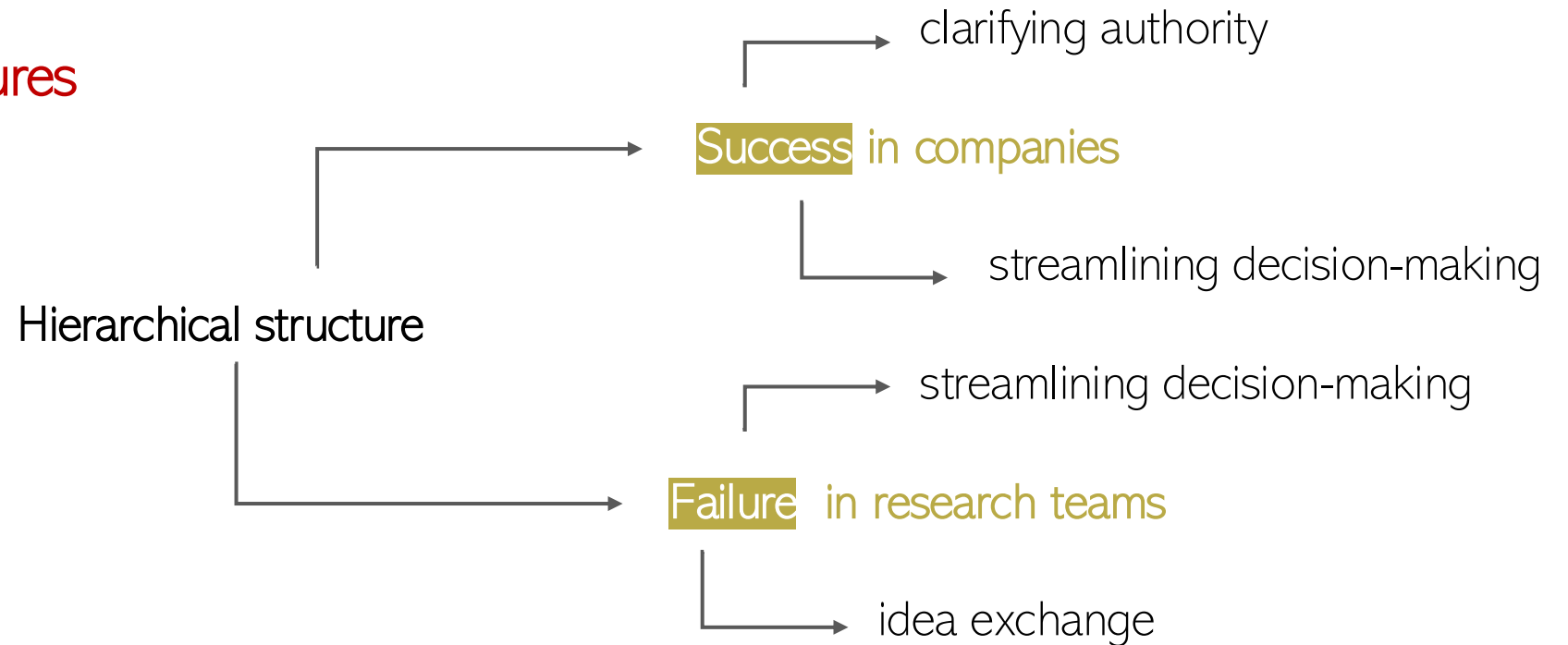
Refining a team

- Improving the effectiveness of initially formed team.
- Enhancing the collaboration among team members.

- Structural adjustment
- Membership substitution

Structural adjustment

Task-dependent structures



[Almutairi et. Al., Human-Computer Interaction for Work (CHIWORK), 2025]

[Zhou et. Al., CHI Conference on Human Factors in Computing Systems, 2018]

[He et. Al., ACM Hum.-Comput. Interact. 1, CSCW, 2017]

Structural adjustment

Feedback

clarifying authority

Failure in research teams

idea exchange

[Almutairi et. Al., Human-Computer Interaction for Work (CHIWORK), 2025]

[Zhou et. Al., CHI Conference on Human Factors in Computing Systems, 2018]

[He et. Al., ACM Hum.-Comput. Interact. 1, CSCW, 2017]

Structural adjustment

DreamTeam, a multi-armed bandit chatbot.

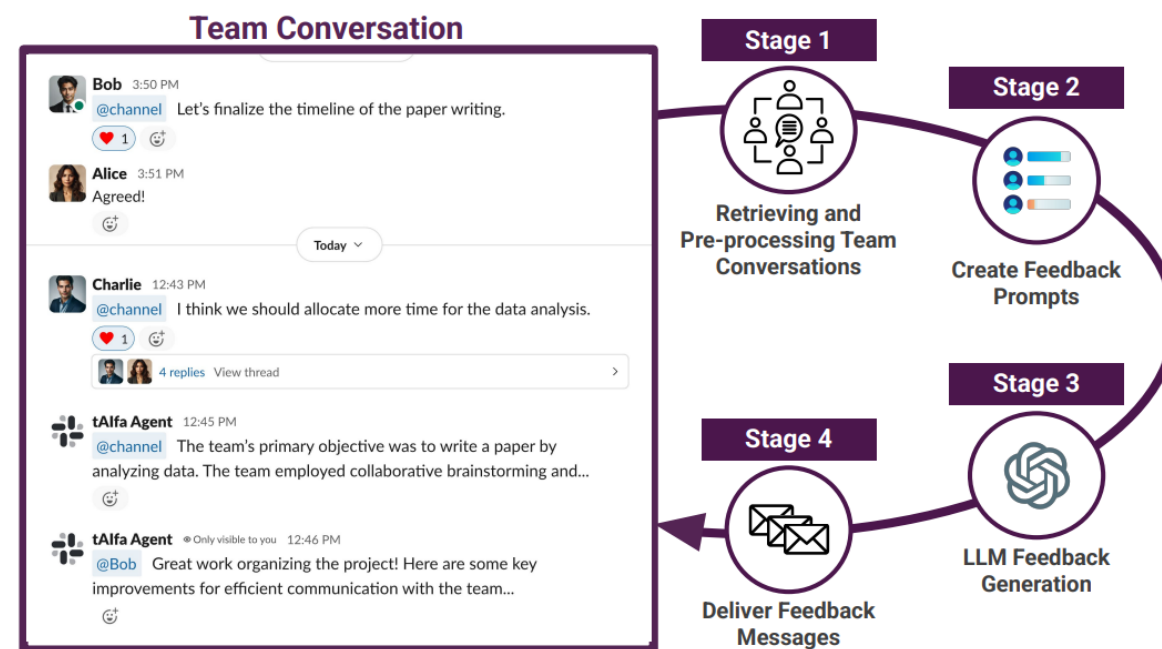
- DreamTeam takes in a set of dimensions representing team structures.
- The system reacts based on **feedback** from automatically collected **metrics** such as **task performance**, or self-reported metrics such as member ratings on the team's collaboration
- Reward function encapsulates these metrics.
- DreamTeam learns and selects what to explore next based on the reward function.

DIMENSION BANDIT	VALUES ARMS
Hierarchy Early constraint	- None: no structure enforced, teams can do anything - Centralized: elect a leader - Decentralized: majority-led vote to determine responses
Interaction patterns Ongoing constraint	- Emergent: allow patterns to emerge organically - Round-robin: take turns making suggestions - Equally distributed: post in similar quantities as teammates
Norms of Engagement Late constraint	- None: no structure enforced, teams can do anything - Professional: use professional language with each other - Informal: get to know your teammates and add fun to the task
Decision-making Norms Late constraint	- None: no structure enforced, teams can do anything - Divergent: think of diverse ideas - Convergent: generate consensus and use compromise - Informed: make informed and thoughtful judgments - Rapid: make decisions as quickly as possible
Feedback Norms Ongoing constraint	- None: no structure enforced, teams can do anything - Encouraging: give positive encouraging comments to teammates - Critical: critique and play devil's advocate

Structural adjustment

tAlfa, an AI agent that uses LLMs to provide personalized, automated feedback to teams and their members.

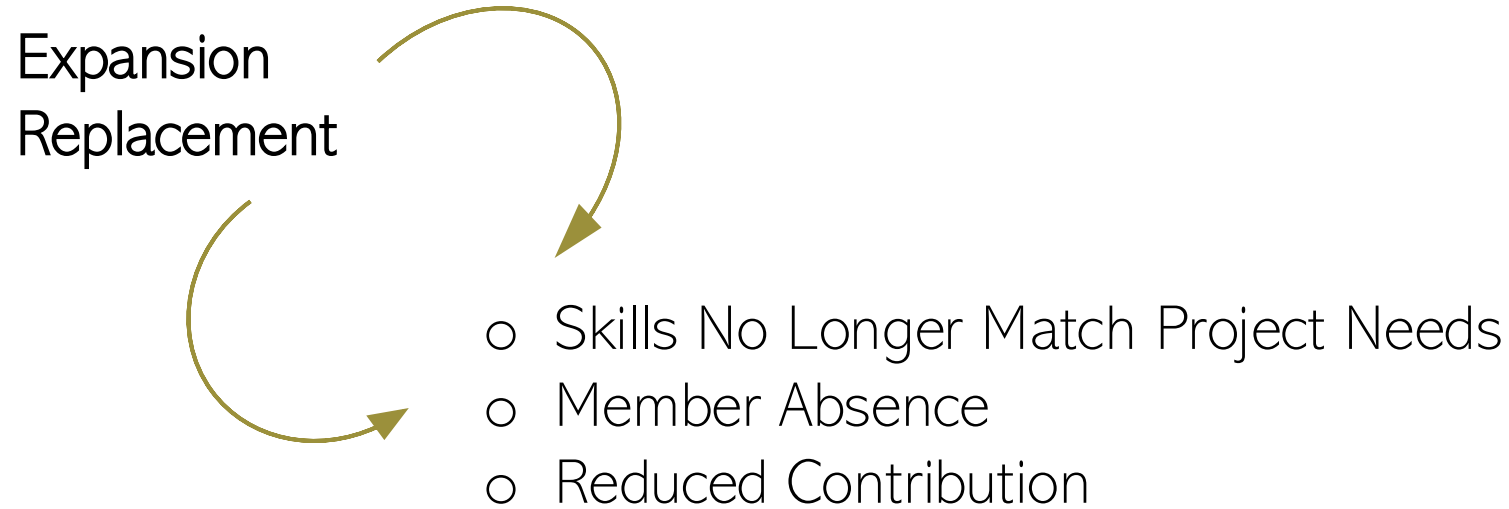
- Retrieve and structure team conversations.
- Evaluate team dynamics using rule-based metrics and LLM analysis.
- Use LLM to generate individual and team feedback messages.
- Deliver feedback privately and as group messages tagging all members.



Evaluation Metrics:

- Average Conversation Duration
- Average Speaker Turn
- Average Word Count
- Self-Reported Measures

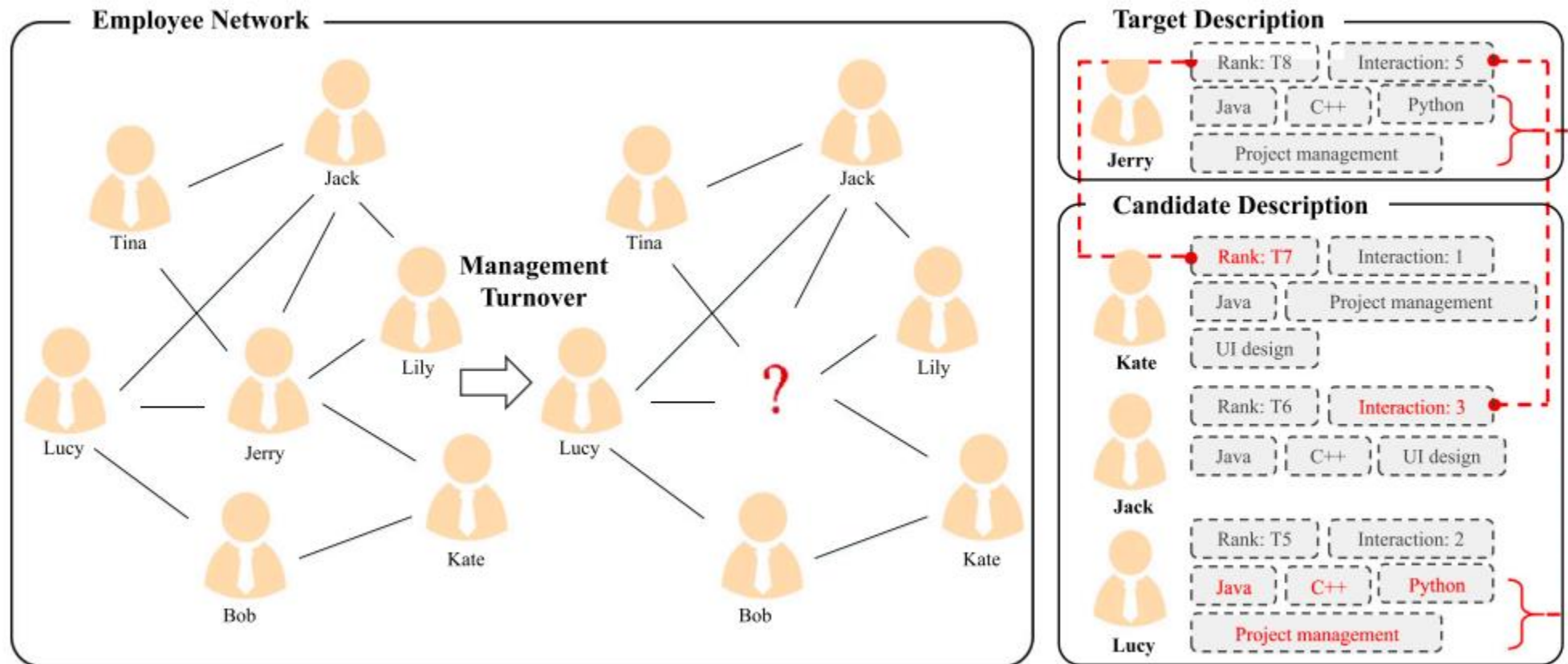
Membership substitution



[Lv et. Al., Decision Support Systems, 2024]

[Zhao et. Al., Springer International Publishing AG, part of Springer Nature, 2018]

Membership substitution



Membership substitution

TECE, Team Expansion in Collaborative Environments

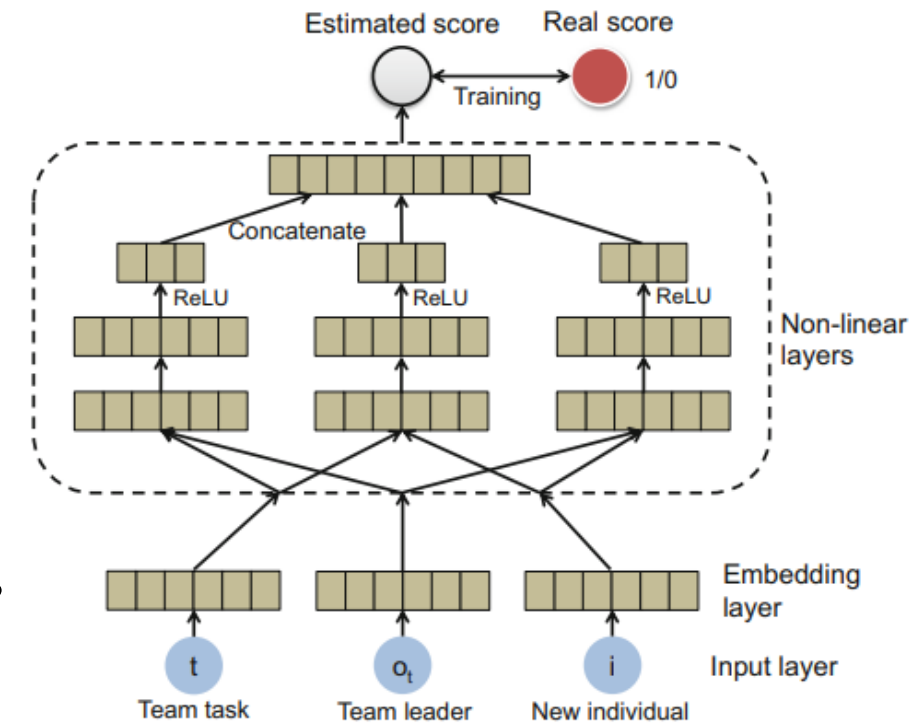
Goal:

Select the most suitable candidate to **expand** a team by jointly considering the **relationships** among the candidate, **team task**, and existing **team members**.

Membership substitution

TECE, predict a unified fitness score

- **Task–Candidate Interaction:** Learns how candidate skills align with task requirements using collaborative filtering.
- **Candidate–Leader Interaction:** Models social compatibility through interactions between the candidate and team leader.
- **Team State Interaction:** Captures current team dynamics via task–leader relationships to find complementary candidates.



Membership substitution

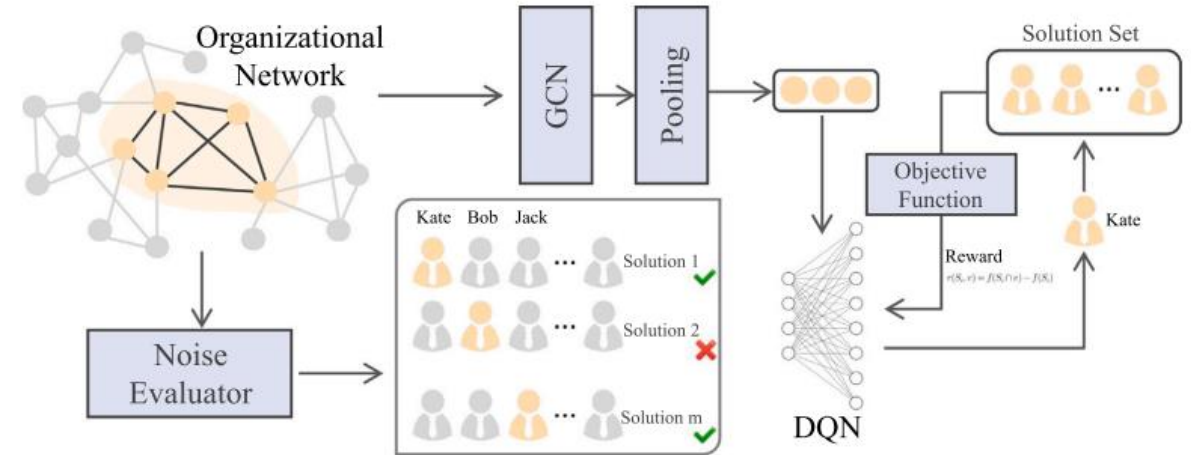
GCO-DQN, Graph Combinatorial Optimization for Deep Quality Network

Goal:

learn member representations using **reinforcement learning** and **graph neural networks** to form teams that satisfy new project requirements while **minimizing disruption** from **reallocating** members from their previous teams.

Membership substitution

GCO-DQN,



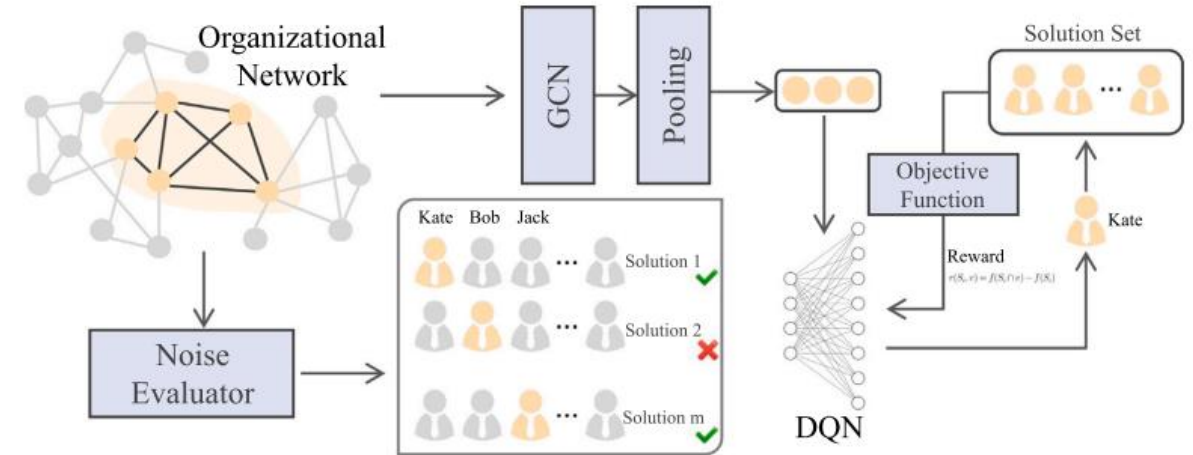
➤ Problem Setup

Form project teams from an organizational network while meeting skill requirements and minimizing disruption.

Employees are nodes with skills, rank, and communication links.

Membership substitution

GCO-DQN,

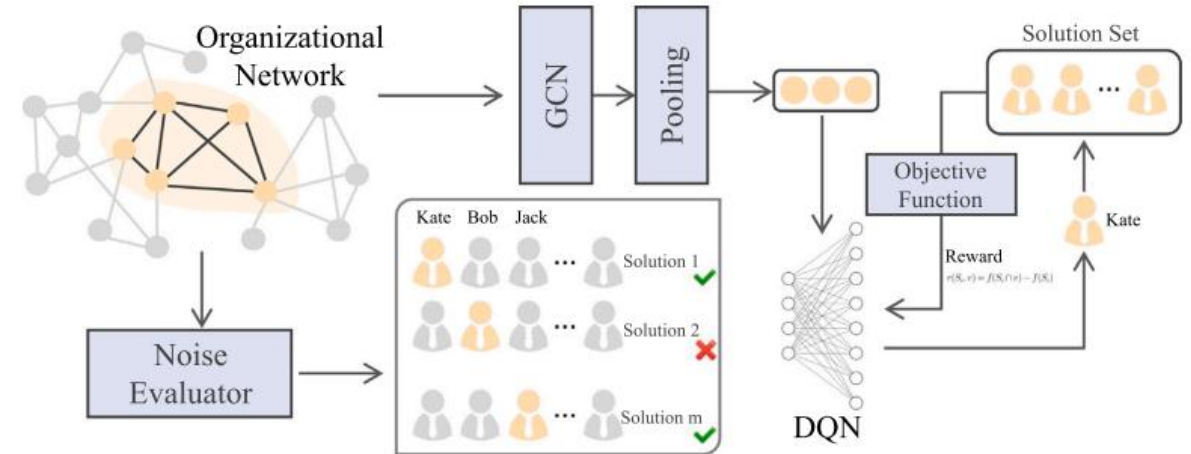


➤ Data Preparation:

Use a probabilistic greedy algorithm to generate multiple candidate team solutions.

Membership substitution

GCO-DQN,



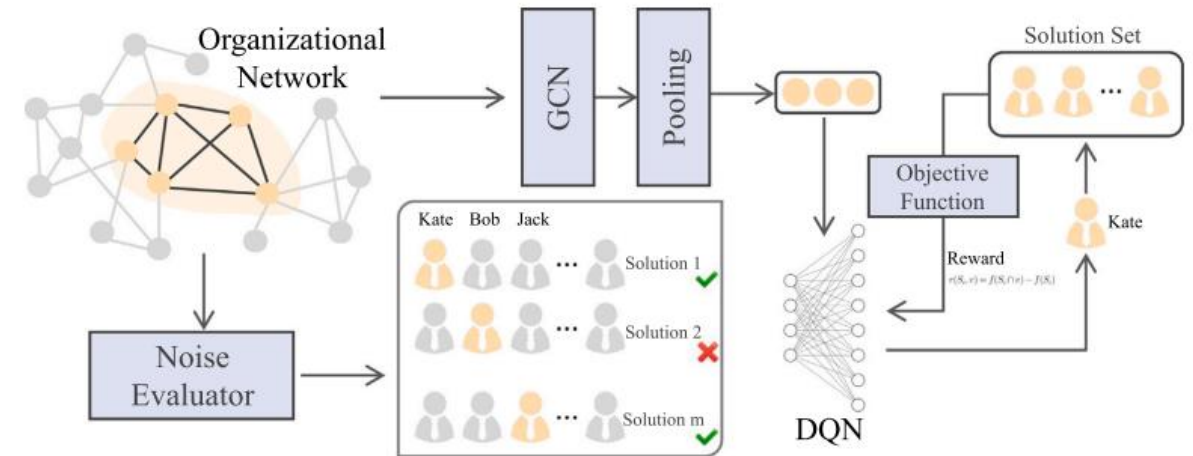
➤ Node Quality Prediction (GNN)

Graph Neural Network evaluates each employee's potential contribution based on skills and network influence.

Filters out less suitable candidates and ranks employees for team selection.

Membership substitution

GCO-DQN,



➤ Team Formation via Reinforcement Learning (GCO-DQN)

- State: current team composition.
- Action: adding an employee.
- Reward: improvement in team effectiveness.

Team Formation → Refinement Techniques

- Forming teams by applying team refinement methods.
- Refinement techniques provide a way to **iteratively** adjust team composition toward higher performance teams.

Team Formation → Refinement Techniques

➤ Forming teams by applying team refinement methods

Reinforcement Learning

Team Formation → Refinement Techniques

Leverage Refinement via **Reinforcement Learning (RL)**:

- Model team formation as a sequential decision-making process.
- Start with an initial team and iteratively **refine** it by adding, removing, or swapping members.
- An RL agent receives **feedback** (reward) based on the improvement in team performance metrics.

Team Formation → Refinement Techniques

Advantages

- Captures **dynamic** and contextual dependencies.
- Adapts through **experience** and **feedback**.

Challenges

- Sparse or delayed rewards make training **unstable**.
- **Limited** applicability in real-world expert domains (e.g., research or organizational teams).

Team Formation → Refinement Techniques

The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI-20)

Refinement-Based Team Formation in Game and Simulated Environments

- **State space:** Current team composition, available candidates, task requirements.
- **Actions:** Propose to add/remove members, accept/reject negotiation offers.
- **Reward function:** Captures team performance, success, fairness, or stability of formed teams.

Google Research Football: A Novel Reinforcement Learning Environment

Karol Kurach,* Anton Raichuk,* Piotr Stańczyk,* Michał Zajac†
Olivier Bachem Lasse Espeholt Carlos Riquelme Damien Vincent
Marcin Michalski Olivier Bousquet Sylvain Gelly
Google Research, Brain Team

Abstract

Recent progress in the field of reinforcement learning has been accelerated by virtual learning environments such as video games, where novel algorithms and ideas can be quickly tested in a safe and reproducible manner. We introduce the *Google Research Football Environment*, a new reinforcement learning environment where agents are trained to play football in an advanced, physics-based 3D simulator. The resulting environment is challenging, easy to use and customize, and it is available under a permissive open-source license. In addition, it provides support for multiplayer and multi-agent experiments. We propose three full-game scenarios of varying difficulty with the *Football Benchmarks* and report baseline results for three commonly used reinforcement algorithms (IMPALA, PPO, and Ape-X DQN). We also provide a diverse set of simpler scenarios with the *Football Academy* and showcase several promising research directions.



Figure 1: The *Google Research Football Environment* (github.com/google-research/football) provides a novel reinforcement learning environment where agents are trained to play football in an advanced, physics based 3D simulation.

Outline

I) Introduction and Background

II) Pioneering Techniques

III) Learning-based Heuristics

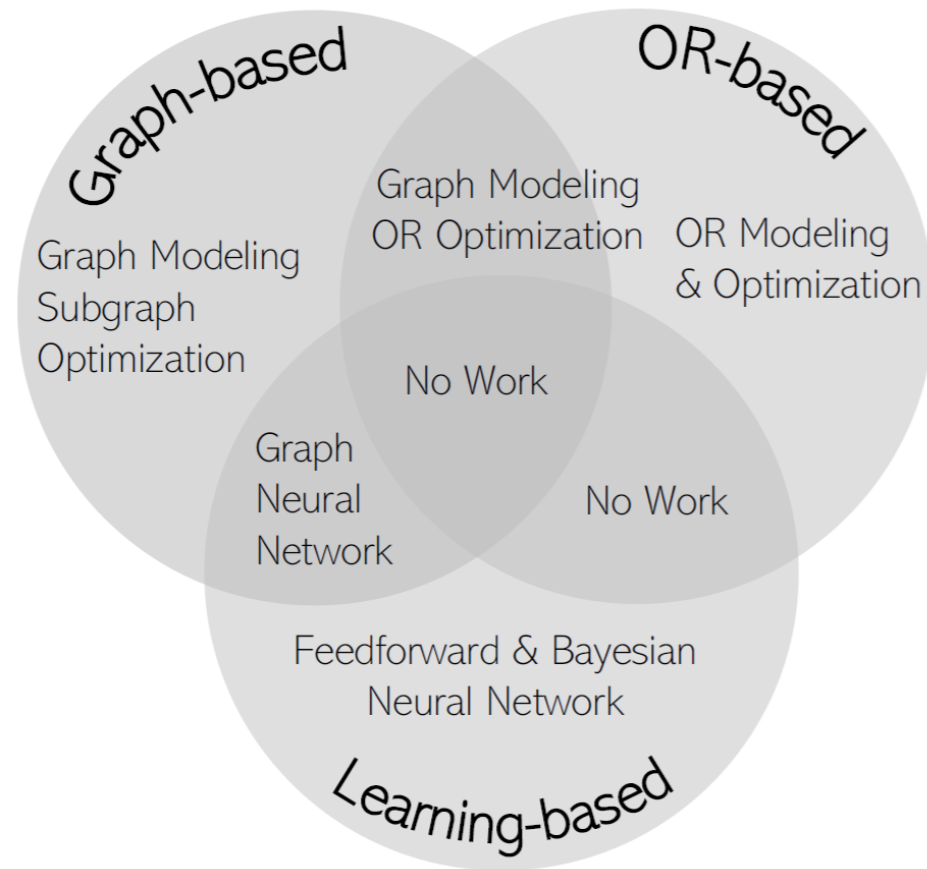
IV) Team Refinement

V) Challenges and New Perspectives

VI) Applications

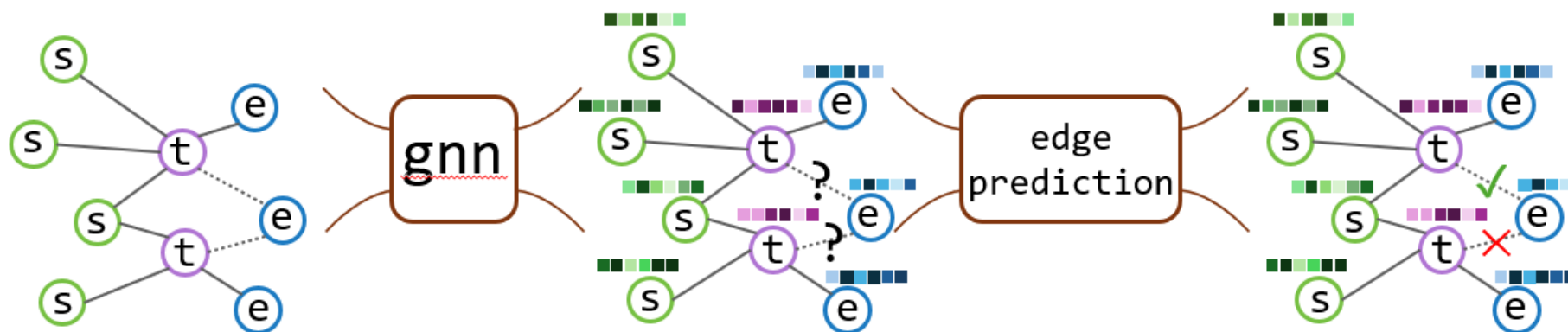
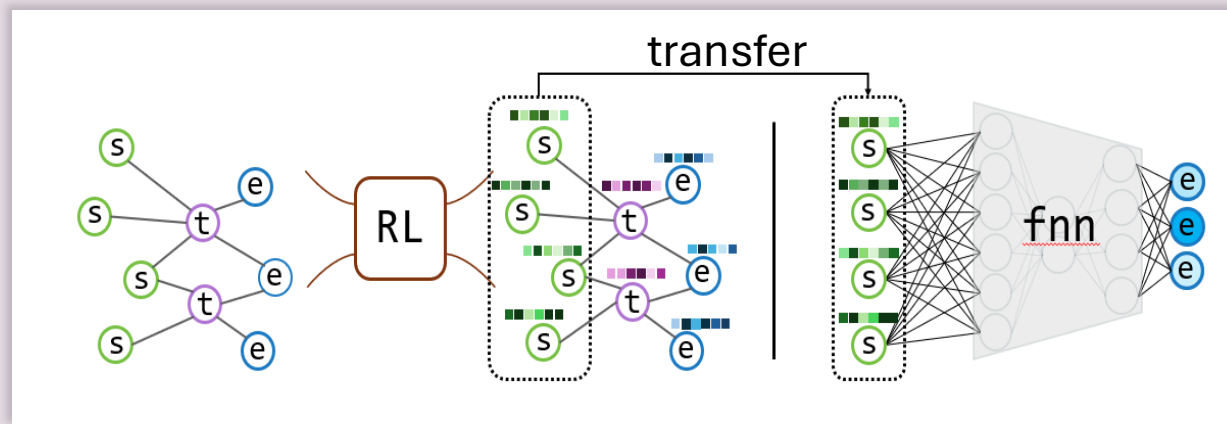
Hands-on: OpeNTF

Immediate Direction



End-to-End Approach

Transfer learning vs. End-to-End



End-to-End Approach

		dblp								imdb							
		%pr		%rec		%ndcg		%map		%pr		%rec		%ndcg		%map	
		@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10
e2e	gs	14.37	09.24	54.00	67.79	33.21	38.39	25.18	28.10	15.39	09.16	49.73	57.44	33.70	36.78	27.00	29.03
	gin	14.47	09.07	54.68	66.95	31.50	36.12	22.74	25.36	14.08	08.91	45.34	56.49	28.49	32.79	21.76	24.30
	gine	07.66	07.65	21.56	44.17	16.50	24.74	12.67	16.18	19.33	12.21	53.14	66.25	38.80	43.86	31.13	34.07
	gat	14.33	09.46	53.81	68.86	38.11	43.78	31.44	34.65	14.66	09.17	47.44	57.47	33.45	37.40	27.17	29.68
	gatv2	14.03	09.48	52.73	69.01	37.87	43.98	31.43	34.86	14.49	09.09	46.72	57.18	33.81	37.88	27.86	30.39
	han	28.97	15.17	83.19	86.38	70.42	71.74	64.28	65.27	23.37	15.42	57.82	74.15	45.58	52.38	37.42	42.21
t-fnn	mhot	00.70	00.65	01.07	01.96	00.96	01.37	00.53	00.66	00.75	00.70	00.86	01.59	00.84	01.19	00.41	00.52
	d2v	00.83	00.72	01.27	02.21	01.18	01.62	00.72	00.88	00.84	00.76	00.93	01.72	00.95	01.32	00.45	00.57
	m2v	00.82	00.70	01.24	02.13	01.13	01.55	00.66	00.82	00.84	00.77	00.95	01.76	00.95	01.33	00.46	00.57
	lant	00.90	00.83	01.35	02.48	01.17	01.68	00.72	00.88	00.83	00.68	00.81	01.52	00.85	01.18	00.40	00.50
	gs	00.87	00.74	01.30	02.24	01.26	01.63	00.60	00.76	00.83	00.73	00.91	01.64	00.95	01.27	00.44	00.55
	gin	00.95	00.78	01.43	02.35	01.37	01.57	00.59	00.76	00.76	00.68	00.84	01.56	00.87	01.19	00.40	00.51
	gine	01.02	00.81	01.52	02.49	01.35	01.80	00.76	00.92	01.73	01.64	00.82	01.45	00.95	01.14	00.41	00.50
	gat	01.19	00.92	01.78	02.73	01.62	02.08	00.92	01.97	00.92	00.82	01.00	01.83	01.00	01.43	00.50	00.62
	gatv2	01.01	00.83	01.51	02.51	01.41	01.81	00.87	00.97	00.87	00.80	00.96	01.80	00.98	01.37	00.47	00.59
	han	00.93	00.82	01.42	02.49	01.33	01.82	00.80	00.97	00.77	00.72	00.86	01.63	00.85	01.21	00.39	00.50
t-bnn	mhot	00.53	00.49	00.82	01.51	00.73	01.05	00.43	00.54	00.83	00.74	00.95	01.71	00.92	01.28	00.45	00.57
	d2v	00.68	00.61	01.02	01.86	00.95	01.33	00.56	00.69	00.87	00.81	00.99	01.84	00.98	01.38	00.47	00.60
	m2v	00.31	00.32	00.48	00.98	00.42	00.65	00.23	00.30	00.76	00.71	00.83	01.58	00.85	01.20	00.41	00.52
	lant	00.66	00.61	01.01	01.87	00.88	01.28	00.49	00.61	01.01	00.96	01.14	02.18	01.18	01.66	00.58	00.73
	gs	00.71	00.66	01.08	02.00	00.94	01.36	00.52	00.65	00.99	00.90	01.12	02.03	01.12	01.55	00.54	00.67
	gin	00.60	00.52	00.91	01.59	00.84	01.15	00.49	00.60	00.90	00.83	01.04	01.92	01.07	01.47	00.52	00.65
	gine	00.75	00.63	01.12	01.90	01.08	01.44	00.66	00.79	01.01	00.93	01.15	02.14	01.16	01.62	00.58	00.71
	gat	00.66	00.62	00.99	01.86	00.87	01.28	00.49	00.62	00.96	00.86	01.07	01.94	01.10	01.50	00.53	00.66
	gatv2	00.64	00.54	00.96	01.64	00.87	01.19	00.49	00.58	00.93	00.80	01.05	01.84	01.05	01.41	00.51	00.62
	han	00.80	00.71	01.20	02.15	01.09	01.53	00.61	00.76	01.06	00.99	01.19	02.25	01.21	01.70	00.59	00.74

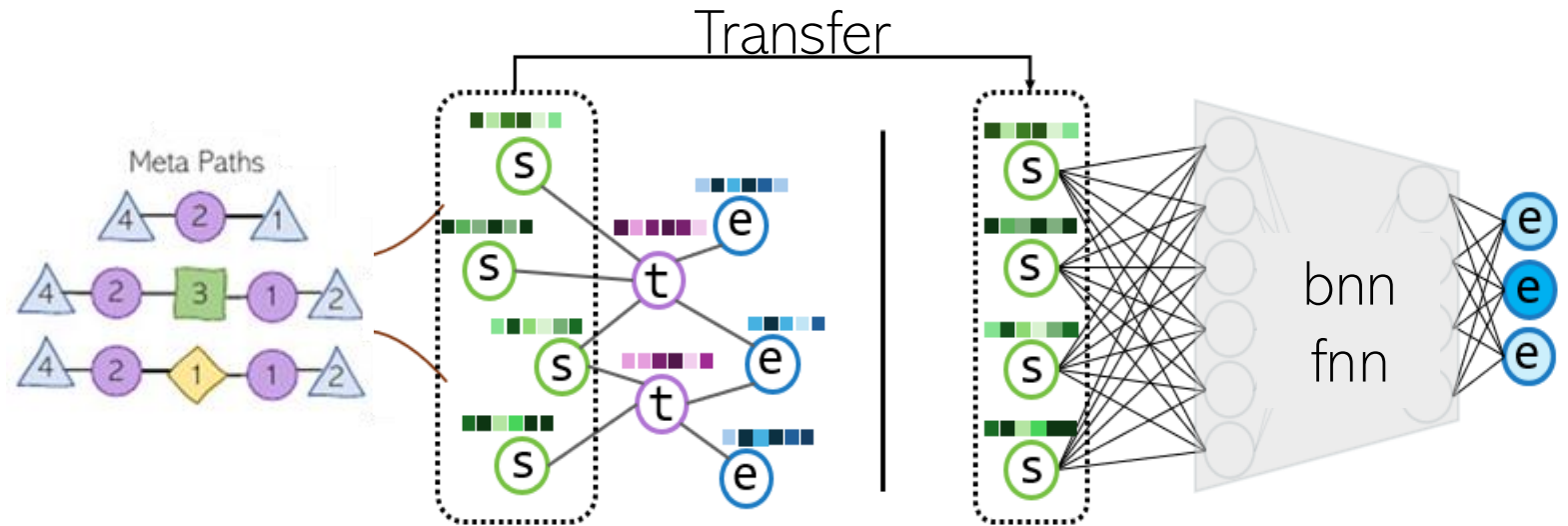
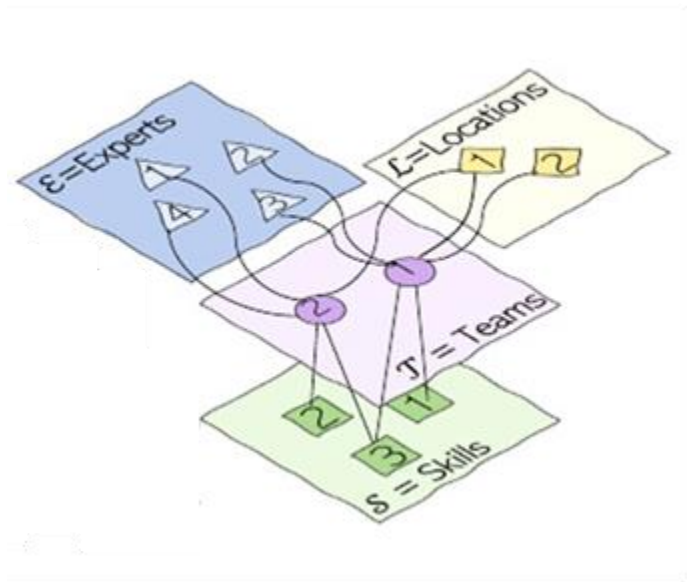
Research Question

Does end-to-end approach outperform transfer learning-based approach for team recommendation?

End-to-End Approach

		dblp								imdb							
		%pr		%rec		%ndcg		%map		%pr		%rec		%ndcg		%map	
		@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10
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	d2v	00.83	00.72	01.27	02.21	01.18	01.62	00.72	00.88	00.84	00.76	00.93	01.72	00.95	01.32	00.45	00.57
	m2v	00.82	00.70	01.24	02.13	01.13	01.55	00.66	00.82	00.84	00.77	00.95	01.76	00.95	01.33	00.46	00.57
	lant	00.90	00.83	01.35	02.48	01.17	01.69	00.67	00.84	00.74	00.68	00.81	01.52	00.85	01.18	00.40	00.50
	gs	00.87	00.74	01.30	02.24	01.26	01.63	00.60	00.76	00.83	00.73	00.91	01.64	00.95	01.27	00.44	00.55
	gin	00.95	00.78	01.43	02.35	01.37	01.57	00.59	00.76	00.76	00.68	00.84	01.56	00.87	01.19	00.40	00.51
	gine	01.02	00.83	01.52	02.49	01.35	01.80	00.76	00.92	00.73	00.64	00.82	01.45	00.85	01.14	00.41	00.50
	gat	01.19	00.92	01.78	02.75	01.62	02.08	00.92	01.09	00.92	00.82	01.00	01.83	01.06	01.43	00.50	00.62
	gatv2	01.01	00.83	01.51	02.51	01.41	01.88	00.80	00.96	00.87	00.80	00.96	01.80	00.98	01.37	00.47	00.59
	han	00.93	00.82	01.42	02.49	01.33	01.82	00.80	00.97	00.77	00.72	00.86	01.63	00.85	01.21	00.39	00.50
t-bnn	mhot	00.53	00.49	00.82	01.51	00.73	01.05	00.43	00.54	00.83	00.74	00.95	01.71	00.92	01.28	00.45	00.57
	d2v	00.68	00.61	01.02	01.86	00.95	01.33	00.56	00.69	00.87	00.81	00.99	01.84	00.98	01.38	00.47	00.60
	m2v	00.31	00.32	00.48	00.98	00.42	00.65	00.23	00.30	00.76	00.71	00.83	01.58	00.85	01.20	00.41	00.52
	lant	00.66	00.61	01.01	01.87	00.88	01.28	00.49	00.61	01.01	00.96	01.14	02.18	01.18	01.66	00.58	00.73
	gs	00.71	00.66	01.08	02.00	00.94	01.36	00.52	00.65	00.99	00.90	01.12	02.03	01.12	01.55	00.54	00.67
	gin	00.60	00.52	00.91	01.59	00.84	01.15	00.49	00.60	00.90	00.83	01.04	01.92	01.07	01.47	00.52	00.65
	gine	00.75	00.63	01.12	01.90	01.08	01.44	00.66	00.79	01.01	00.93	01.15	02.14	01.16	01.62	00.58	00.71
	gat	00.66	00.62	00.99	01.86	00.87	01.28	00.49	00.62	00.96	00.86	01.07	01.94	01.10	01.50	00.53	00.66
	gatv2	00.64	00.54	00.96	01.64	00.87	01.19	00.49	00.58	00.93	00.80	01.05	01.84	01.05	01.41	00.51	00.62
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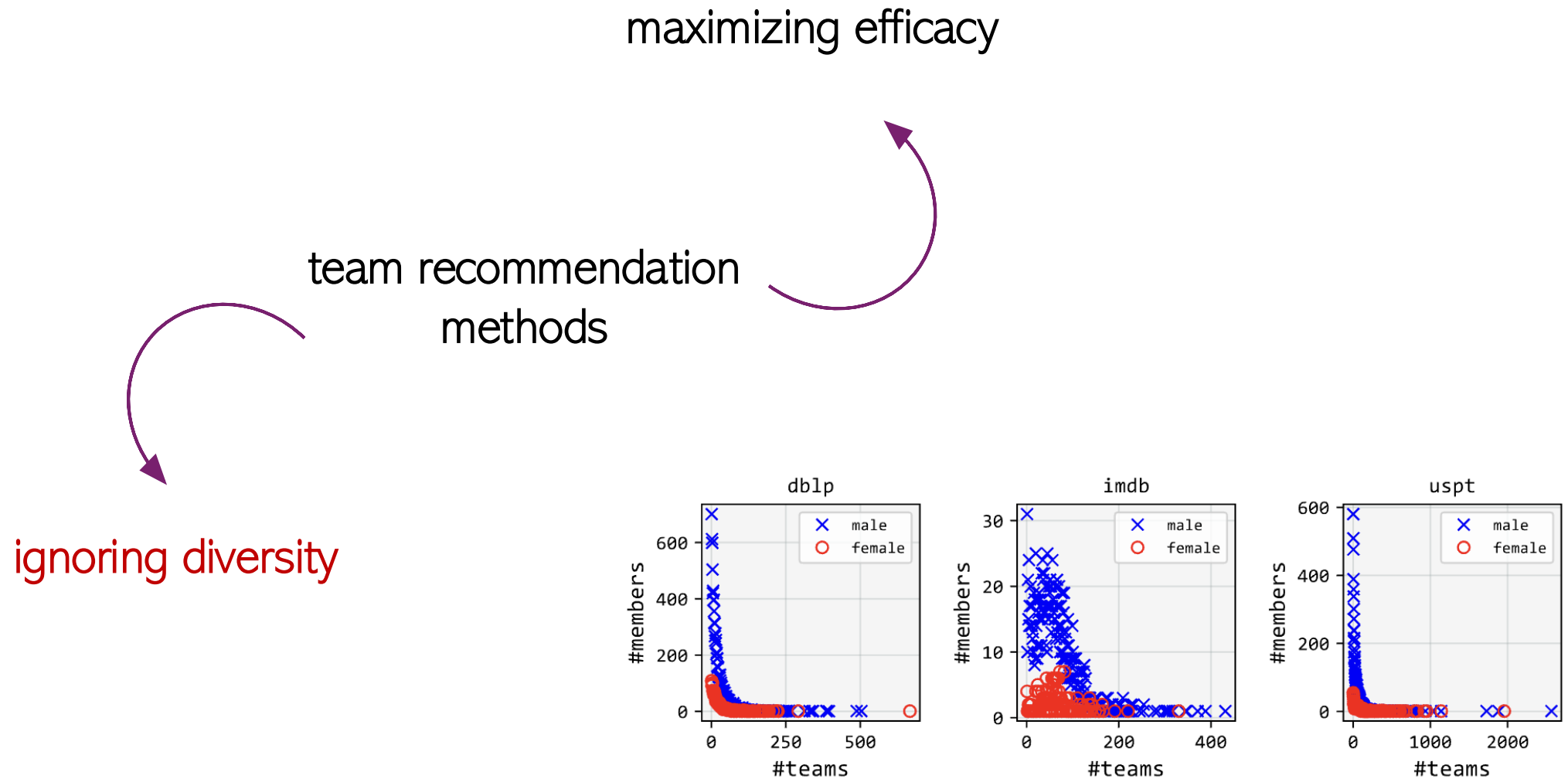
Spatial Team Recommendation



Geolocations only for skill vectors!

[Rad et. al., SIGIR 2021]

Fair and Diverse Team Recommendation



Fair and Diverse Team Recommendation

Fairness in machine learning algorithms guarantees where a disadvantaged group also known as a **protected group**, should be treated similarly to the advantaged group as a whole.

[Altenburger et.al., AAAI Conference on Web and Social Media, 2017]

There is little to no **diversity-aware algorithmic method** that mitigates unfair societal biases in team recommendation models.

Fair and Diverse Team Recommendation

There is little to no **diversity-aware algorithmic method** that mitigates unfair societal biases in team recommendation models.

Notions of **Group** Fairness:

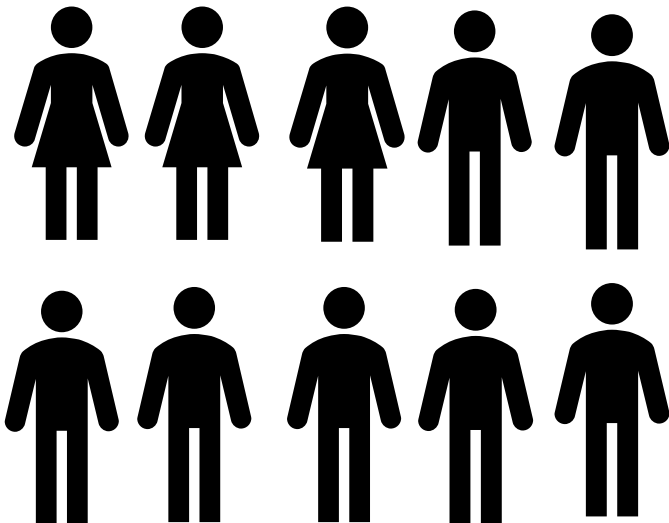
- Demographic Parity
- Equality of Opportunity



Fair and Diverse Team Recommendation

Demographic Parity

Candidate Pool



Demographic Parity

Final Team



- Male: 70%
- Female: 30%

Fair and Diverse Team Recommendation

Demographic Parity :

Enforces the membership in a team to be independent of values of a protected attribute for team members.

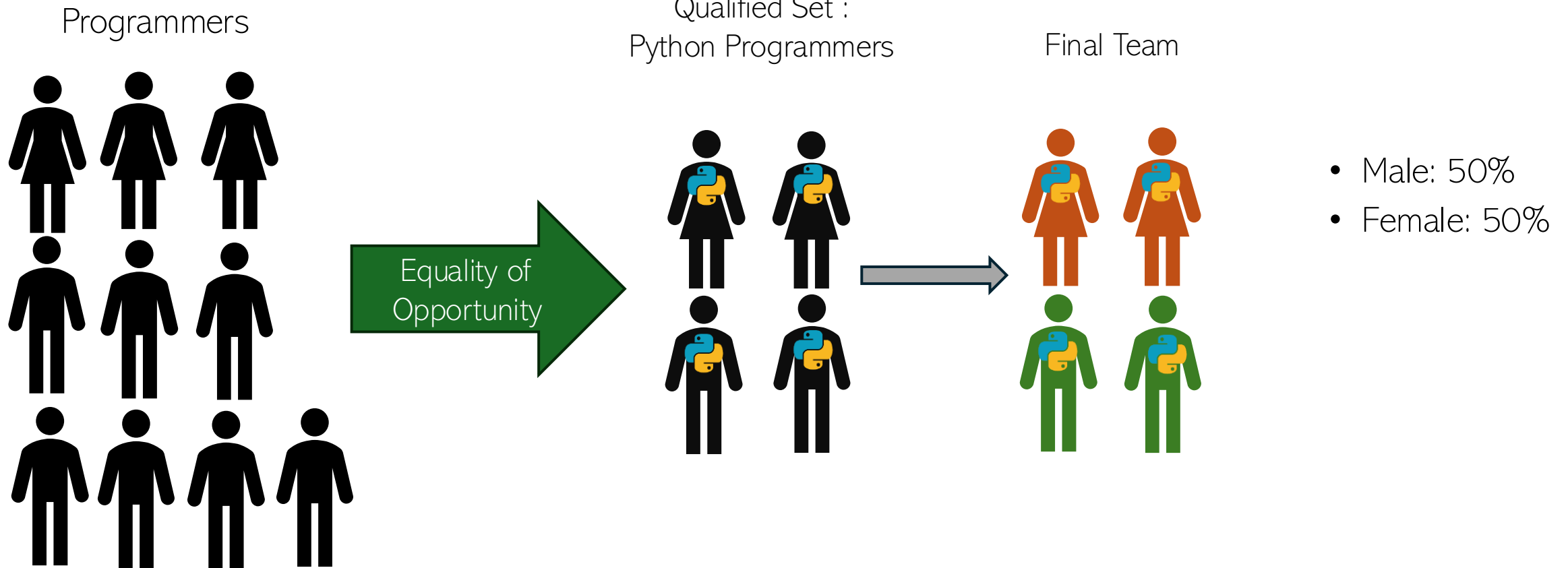
$$[P(e_0 \in T) = P(e'_1 \in T)] \wedge [P(e_0 \notin T) = P(e'_1 \notin T)]$$

Male

Female

Fair and Diverse Team Recommendation

Equality of Opportunity



Fair and Diverse Team Recommendation

Equality of Opportunity:

Enforces the **skilled** membership in a team to be independent of values of a protected attribute for team members.

$$[P(e_0 \in T | e_0, S_{e_0} \cap S \neq \emptyset) = P(e'_1 \in T | e'_1, S_{e'_1} \cap S \neq \emptyset)]$$

Skilled Male

Skilled Female

Fair and Diverse Team Recommendation

Debiasing algorithms can be categorized based on their integration into the machine learning pipeline:

- **Pre-process:** No work, to the best of our knowledge
- **In-process:** Little work, *vivaFemme* [Moasses et.al., BIAS-SIGIR, 2024]
- **Post-process:** Little work, *Adila* [Loghmani et.al., BIAS-ECIR, 2022], [Geyik et.al., KDD, 2019],

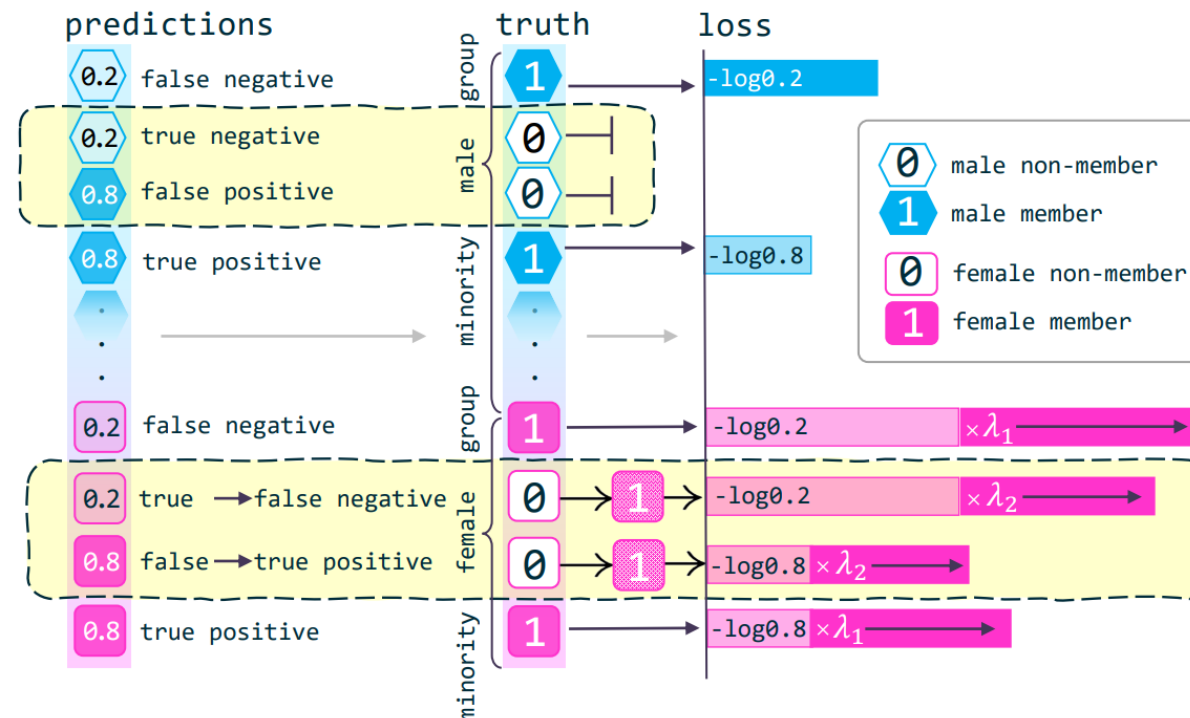
Fair and Diverse Team Recommendation

- **Pre-processing** methods modify data or label through re-sampling heuristics before model training.

Fair and Diverse Team Recommendation

- In-processing** techniques adjust the optimization process of models to balance accuracy and fairness.

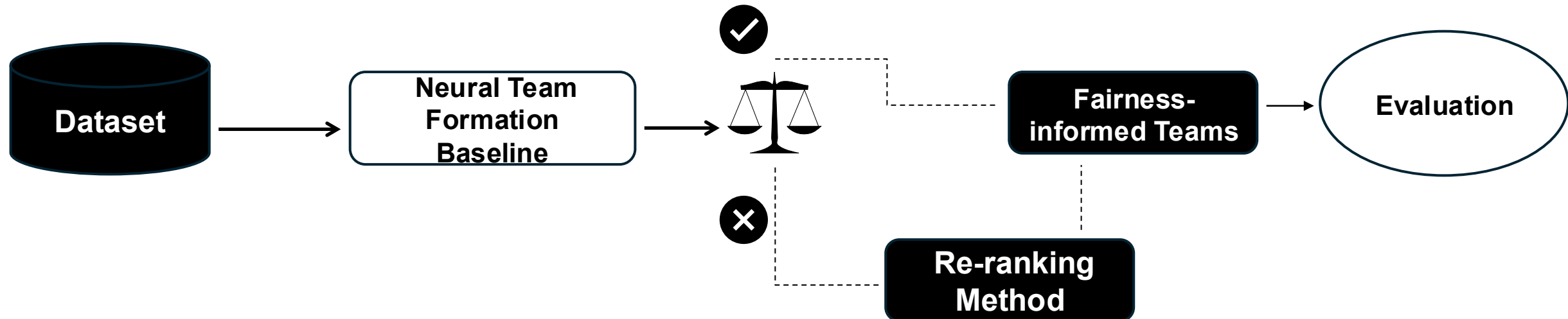
[Moasses et.al., BIAS-SIGIR, 2024]



Fair and Diverse Team Recommendation

- **Post-processing** methods modify model outputs during inference, which may involve altering thresholds, scoring rules, or **reranking** the recommended list of items.

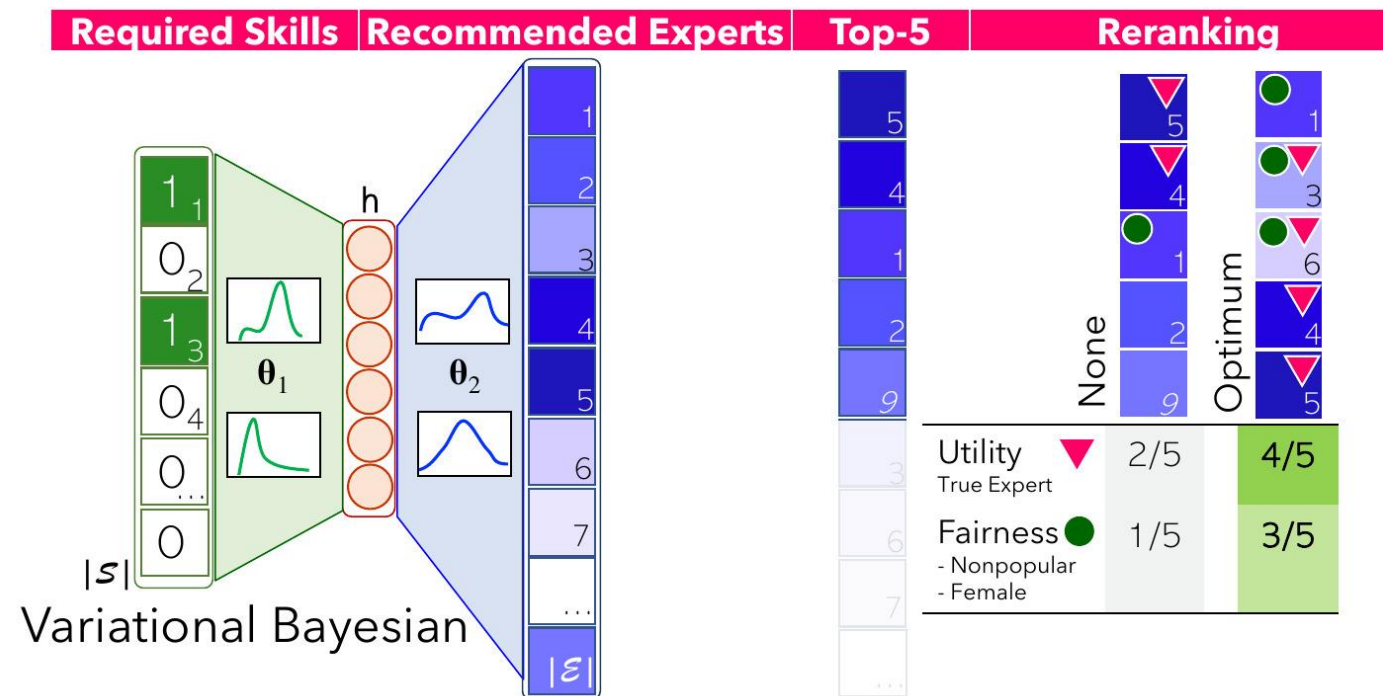
[Loghmani et.al., BIAS-ECIR, 2022], [Geyik et.al., KDD, 2019]



Fair and Diverse Team Recommendation

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[Loghmani et.al., BIAS-ECIR, 2022], [Geyik et.al., KDD, 2019]





Multi-Objective Optimization

Outline

I) Introduction and Background

II) Pioneering Techniques

III) Learning-based Heuristics

IV) Challenges and New Perspectives

V) Team Refinement

VI) Applications

Hands-on: OpeNTF

Applications

- Group Learning
- Reviewer Assignment
- Palliative Care

Applications

- Group Learning
- Reviewer Assignment
- Palliative Care

Group Learning

Why group learning

Group Learning

Why group learning

Active Engagement:

Working in groups helps students engage deeply with the material and share their knowledge

Group Learning

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Accountability:

Teamwork motivates students to enhance problem solving skills, stay on track and meet deadlines.

Group Learning

Why group learning

Active Engagement:

Working in groups helps students engage deeply with the material and share their knowledge

Accountability:

Teamwork motivates students to enhance problem solving skills, stay on track and meet deadlines.

Essential Skills:

Develops time management and communication skills for future success.

Group Learning

Not all student groups work well.

Group Learning

Not all student groups work well. **Why?**

Group Learning

Not all student groups work well. **Why?**

Groups are not **systematically** formed.

Group Learning

Team recommendation approaches in the classroom

- Average grade

Group Learning

Team recommendation approaches in the classroom

- Average grade
- Availability for work

Group Learning

Team recommendation approaches in the classroom

- Average grade
- Availability for work
- Personality and behavior

Group Learning

Prior Methods

Group Learning

Prior Methods

- Integer linear programming (ILP)

Group Learning

Prior Methods

- Integer linear programming (ILP)
- Genetic Algorithm

Group Learning

Prior Methods

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Group Learning

Prior Methods

- Integer linear programming (ILP)
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 - Task assignment to agents based on their capabilities [Crawford et al, AAMAS 2016]

Group Learning

Prior Methods

- Integer linear programming (ILP)
- Genetic Algorithm
 - Resource allocation [Vecina et al, HAIS 2022]
 - Matching of students to supervisors [Sanchez et al, Appl. Soft Comput. 2019]
 - Task assignment to agents based on their capabilities [Crawford et al, AAMAS 2016]
 - Project allocation to individuals according to their skills [Camelo et al, Comput. Oper. Res., 2021]

Applications

- Group Learning
- Reviewer Assignment
- Palliative Care

Reviewer Assignment

Background

Reviewer Assignment

Background

- Peer review is a critical part of the **academic publishing process** and is closely tied to the quality and integrity of academic journals.

[Mandviwalla et. Al., Decision Support Systems, 2008]

[Zhang et. Al., Information Processing and Management, 2023]

Reviewer Assignment

Background

- Peer review is a critical part of the **academic publishing process** and is closely tied to the quality and integrity of academic journals.
- Assigning reviewers to a manuscript is crucial for determining the **quality** and **suitability** of **submitted manuscript**.

[Mandviwalla et. Al., Decision Support Systems, 2008]

[Zhang et. Al., Information Processing and Management, 2023]

Reviewer Assignment

Approaches

Reviewer Assignment Approaches

Reviewer-paper matching degree

Reviewer Assignment Approaches

Reviewer-paper matching degree

- Manual methods such as asking reviewers to bid on papers. [Shah et. Al., Journal of Machine Learning Research, 2018]

Reviewer Assignment Approaches

Reviewer-paper matching based

- Text information-based method to compared the similarity between potential reviewers and manuscripts. [yang et. Al., Applied Soft Computing, 2020]

Reviewer Assignment Approaches

Retrieval-based reviewer assignment

Reviewer Assignment Approaches

Retrieval-based reviewer assignment

- Developed a **recommendation framework** that employs information retrieval techniques to collate publications, keywords, and abstracts by integrating **cosine similarity** assessments between keywords and full-text indices to generate final recommendations. [Arabzadeh et. Al., CIKM, 2024]

Applications

- Group Learning
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- Palliative Care

Palliative Care

Background

- Palliative care is caring holistically for **patients** and families to improve their **quality of life**.

[Manalu et. Al., International Journal of Medical and Health Sciences, 2022]

[Selvarajah et. Al., Innovations in Intelligent Systems and Applications (INISTA), 2018]

Palliative Care

Background

- Palliative care is caring holistically for **patients** and families to improve their **quality of life**.
- It provides an active and comprehensive integrated **health** approach, which is a multidisciplinary approach that **integrates** doctors, nurses, physiotherapists, psychologists, nutritionists and other professions as needed.

[Manalu et. Al., International Journal of Medical and Health Sciences, 2022]

[Selvarajah et. Al., Innovations in Intelligent Systems and Applications (INISTA), 2018]

Palliative Care

Background

- The care system takes a [team-based](#) approach to address the needs of patients and their families.

[Manalu et. Al., International Journal of Medical and Health Sciences, 2022]

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Palliative Care

Background

- The care system takes a **team-based** approach to address the needs of patients and their families.

Challenges:

- Each care provider has limited capabilities and can provide special type of services.
- Several barriers such as geographical distance, communication costs, time availability, etc.

[Manalu et. Al., International Journal of Medical and Health Sciences, 2022]

[Selvarajah et. Al., Innovations in Intelligent Systems and Applications (INISTA), 2018]

Palliative Care as a Team Recommendation Problem

- Care providers as experts.
- Services they can provide as skills.

[Manalu et. Al., International Journal of Medical and Health Sciences, 2022]

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Palliative Care as a Team Recommendation Problem

- Care providers as experts.
- Services they can provide as skills.
- Based on the structure of the network, and relationship among the care providers in the network, the best team of support/care can be identified. (efficient team)

Palliative Care as a Team Recommendation Problem

Mapping the Care Network:

- The care network is mapped to a weighted graph, considering factors such as distance, communication, and contact costs.

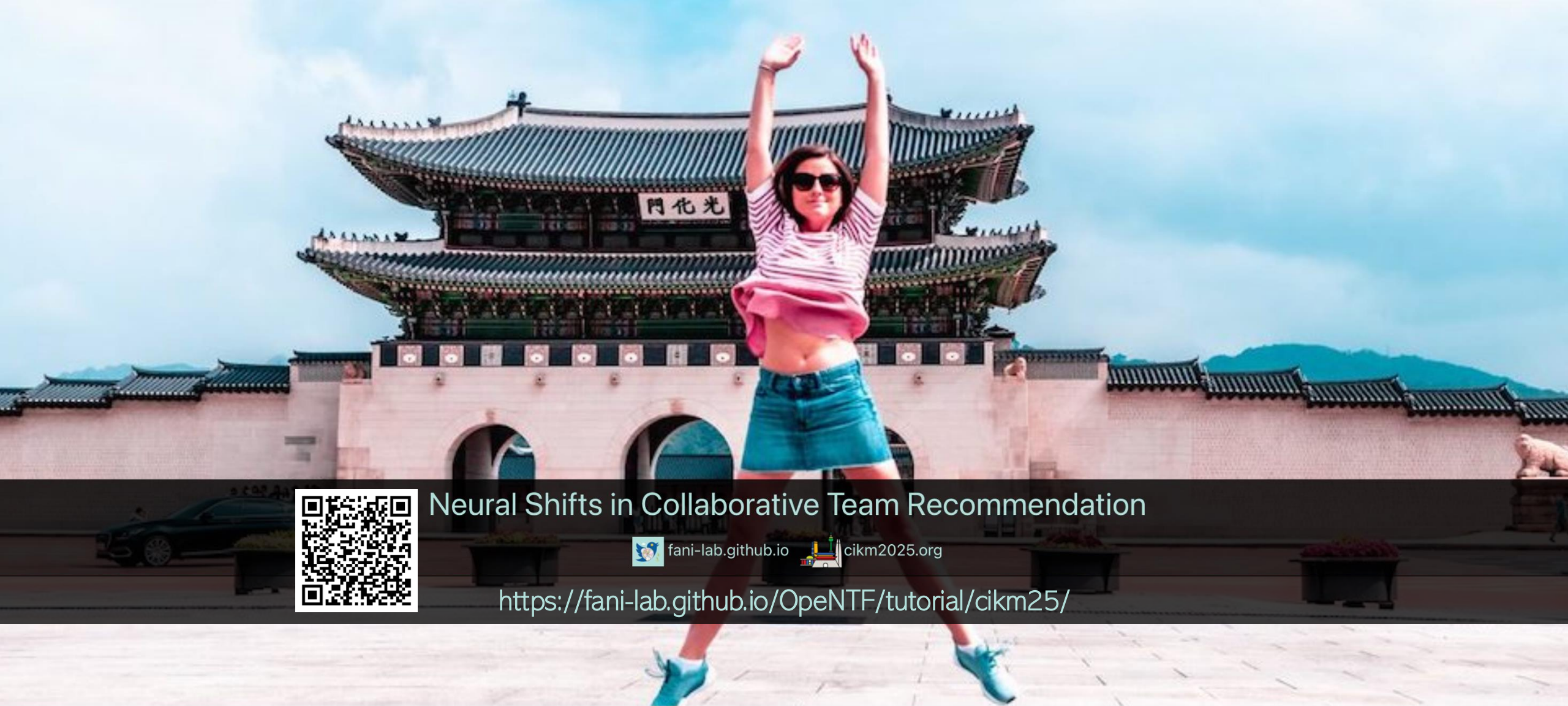
[Manalu et. Al., International Journal of Medical and Health Sciences, 2022]

[Selvarajah et. Al., Innovations in Intelligent Systems and Applications (INISTA), 2018]

Palliative Care Approaches

Evolutionary Modelling

- Cultural Algorithm [Selvarajah et. Al., Innovations in Intelligent Systems and Applications (INISTA), 2018]



Neural Shifts in Collaborative Team Recommendation



fani-lab.github.io



cikm2025.org

<https://fani-lab.github.io/OpeNTF/tutorial/cikm25/>

TIME AND LOCATION

ABSTRACT

TARGET AUDIENCE

OUTLINE

PIONEERING TECHNIQUES

TIME AND LOCATION

Half-day, Morning, 9:00 AM - 12:30 PM (GMT/UTC +9)

Monday, November 10, 2025

COEX, Gangnam, Seoul, South Korea

Prev Tutorials at [UMAP24](#), [SIGIR-AP24](#) and [WSDM25](#) ◀ [\[Full Outline\]](#) [\[Slides\]](#) [\[Recording\]](#)